

Service-oriented Software Engineering (SoSE) PART III

Developing SOAP Web service with Apache CXF

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Roadmap

- What is JAX-WS? (a brief intro)
- JAX-WS high-level Architecture
- JAX-WS Endpoint
- Approaches for creating Web Services
- Apache CXF alone
- Apache CXF with Tomcat

What is JAX-WS? (a brief intro)

Jakarta API for XML Web Services (JAX-WS) is a **specification** of Java programming language API for creating web services, **particularly SOAP services**

JAX-WS is one of the Java XML programming APIs

It is **part** of the **Java EE platform**

- Indeed, **starting from Java 6** it is also part of the Java Standard Edition

Developers do not need to generate or parse SOAP messages

- **runtime environment creates** and **converts SOAP message calls**

JAX-WS **does not prevent interoperability**

- JAX-WS-based client can access a WS realized with a different technology, and vice-versa

https://en.wikipedia.org/wiki/Java_API_for_XML_Web_Services

Communication between a JAX-WS Web Service and a Client



The notions of **endpoint** and **JAX-WS message passing** are needed here...

→ next two slides

JAX-WS Endpoint

To better understand the concept of **JAX-WS Endpoint**, we need to understand the two terms

- **SEI** (Service Endpoint Interface)
- **SIB** (Service Implementation Bean)

Remember that best practices suggest to

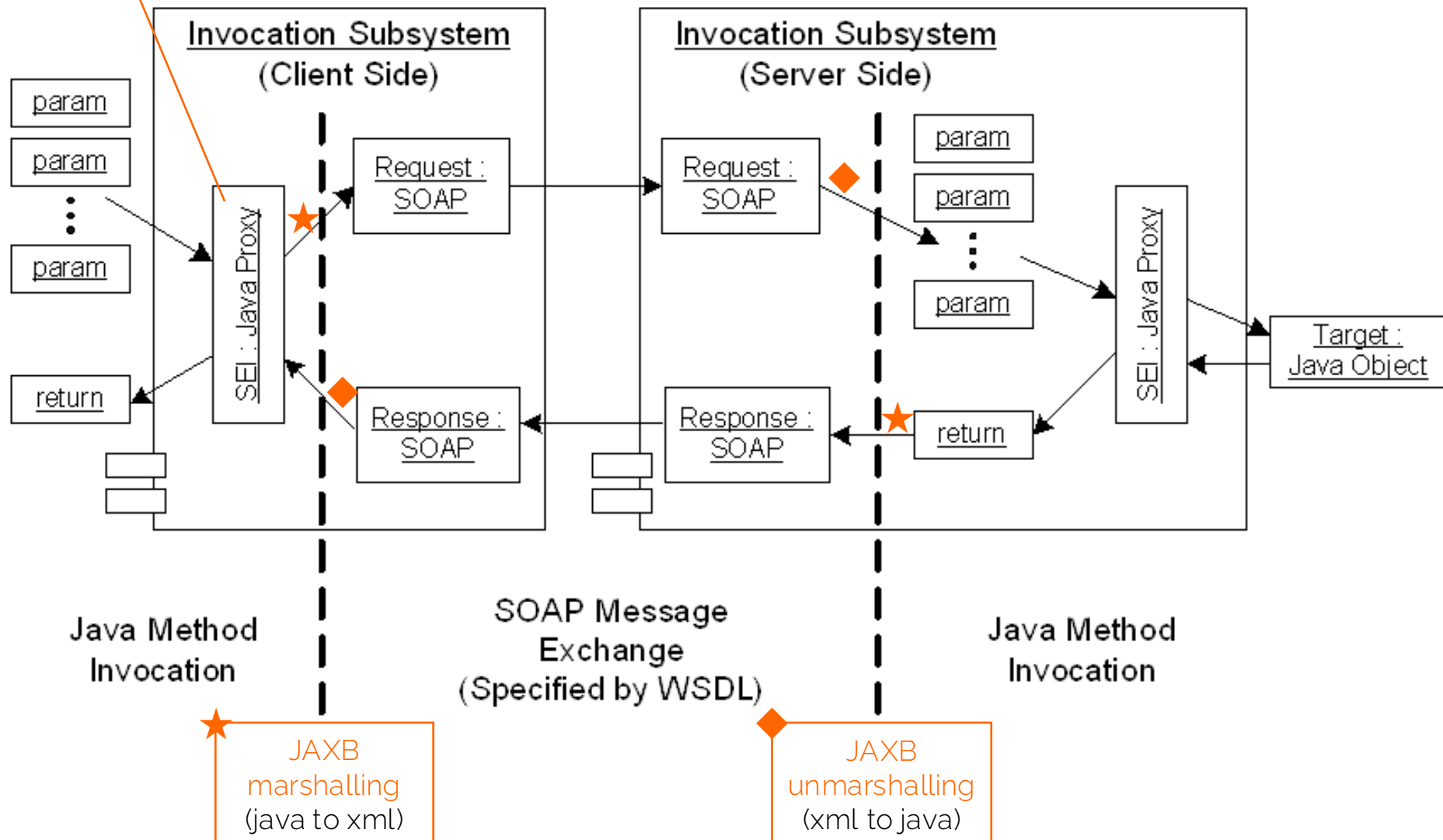
- First, **define an interface** that declares all the methods to be exposed as a Web Service
- Then, **implement** the declared methods

In order to implement a SOAP-based WS by using the JAX-WS API's, up to now, we have defined an **interface** and an **"implementor" class** for it

- The **interface** (annotated with `@WebService`) is an **SEI** and the implementation class (also annotated with `@WebService`) is an **SIB**
- JAX-WS services that do not have an associated SEI are regarded as having an **implicit SEI**
- The **only methods on a JAX-WS** service endpoint that can be invoked by a client are the **business methods defined in the explicit or implicit SEI**

JAX-WS Message Passing

Service Endpoint Interface (SEI)



Approaches

Bottom-up or Code First (Java → WSDL)

- It is about creating web services by starting from the Java classes, annotating them, and generating the corresponding WSDL

Top-Down or Contract First Approach (WSDL → Java)

- It is about creating web services by starting from the WSDL and generating the corresponding Java classes (skeleton classes)

Hybrid

- Start with code first approach by creating Java classes and thereby generating WSDL files. Customize the generated WSDL files, then WSDL -> Java, or viceversa

Approaches

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Advantages of Apache CXF

- Compliant **JAX-WS** implementations (**SOAP** service)
- Compliant **JAX-RS** implementations (**REST** service)
- Supports for **WS-Addressing**, **WS-Policy**, **WS-RM**, **WS-Security** and **WS-I BasicProfile**
- Better integration with **Spring** and **other frameworks**
- Abundant **supporting toolkits**, e.g., Java to WSDL, WSDL to Java, XSD to WSDL, WSDL to XML, WSDL to SOAP, WSDL to Service

From legacy model to modularity (from previous slides)

Starting from Java 11, JAX-WS (and the Endpoint class) was removed from the JDK, requiring external libraries or application servers to develop SOAP services

- Java 9 → deprecated Java 11 → removed

SOAP in Java today is no longer “built-in”, while REST is typically implemented using frameworks such as Spring Boot

Oracle removed the following from the JDK:

- JAX-WS, JAXB, CORBA

Reasons:

- to reduce the size of the JDK
- to separate enterprise technologies from the core platform
- to promote modularity and the use of external libraries

Transition from Java “batteries included” (legacy model) to modular Java + an ecosystem of external libraries (modern model)

Ingredients

We will reuse the same example from the previous slides (using the Endpoint class), but this time with Apache CXF, since, unlike JDK 8, JDK 21 alone is no longer sufficient

Java SE Development Kit 21

<https://www.oracle.com/it/java/technologies/downloads/#java21>

Apache CXF™ (v.4.1.5) binary distribution - zip or tar.gz

<http://cxf.apache.org/download.html>

No Eclipse, no Maven, and no application server will be used

Command line, simply... 😊

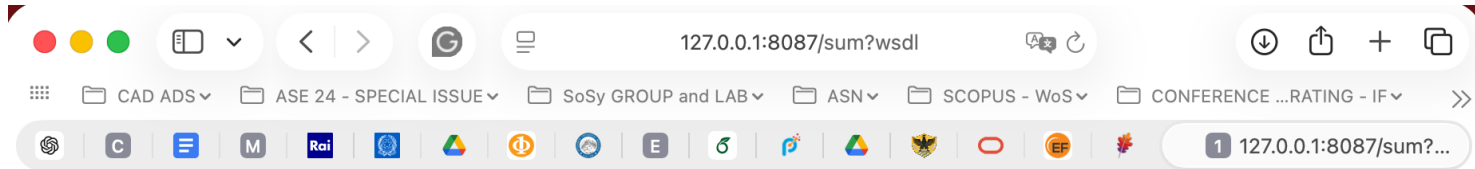
Fundamental rather than simple

The screenshot shows a macOS file explorer window titled "SumSOAPServiceNewNoMavenNoTomcatCLine". The left sidebar shows a tree view with "java" selected, containing "apache", "command-line-examples", "eclipse-workspace", and "spring-workspace". The main pane shows the contents of "SumSOAPServiceNewNoMavenNoTomcatCLine", including "apache-cxf-4.1.5", "build", "lib", "src", "build.sh", "cxf.tar.gz", "fetch-cxf.sh", "README.txt", and "run.sh".

Below the file explorer is a terminal window titled "SumSOAPServiceNewNoMavenNoTomcatCLine — bash — 90x21". The terminal output is as follows:

```
[bash-3.2$  
[bash-3.2$ pwd  
/Applications/java/command-line-examples/SumSOAPServiceNewNoMavenNoTomcatCLine  
bash-3.2$  
bash-3.2$ ./fetch-cxf.sh  
Downloading Apache CXF 4.1.5...  
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current  
     Dload  Upload   Total   Spent    Left   Speed  
100 63.2M  100 63.2M    0     0  2311k      0  0:00:28  0:00:28 --:--:-- 2764k  
[Extracting...  
[Copying libraries...  
Done. Libraries copied to lib/  
bash-3.2$  
bash-3.2$ ./build.sh  
  
You are using /Library/Java/JavaVirtualMachines/jdk-21.jdk/Contents/Home  
[JAVA_HOME = /Library/Java/JavaVirtualMachines/jdk-21.jdk/Contents/Home  
  
Compiled. Classes are in ./build  
bash-3.2$  
bash-3.2$ ./run.sh  
  
...  
INFO: Creating Service {http://docs.oasis-open.org/ws-dd/ns/discovery/2009/01}  
DiscoveryProxy from class org.apache.cxf.jaxws.support.DummyImpl  
Service published at: http://127.0.0.1:8087/sum  
WSDL: http://127.0.0.1:8087/sum?wsdl
```

Our service is up and running



This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8"?>
<wsdl:definitions xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:wsdl="http://schemas.xmlsoap.org/wsdl/" xmlns:tns="http://sum.simple.sose.univaq.it/"
  xmlns:soap="http://schemas.xmlsoap.org/wsdl/soap/" xmlns:ns1="http://schemas.xmlsoap.org/soap/http"
  name="SumImplService" targetNamespace="http://sum.simple.sose.univaq.it/">
  <wsdl:types>
    <xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema"
      xmlns:tns="http://sum.simple.sose.univaq.it/" elementFormDefault="unqualified"
      targetNamespace="http://sum.simple.sose.univaq.it/" version="1.0">
      <xs:element name="sum" type="tns:sum"/>
      <xs:element name="sumResponse" type="tns:sumResponse"/>
      <xs:complexType name="sum">
        <xs:sequence>
          <xs:element name="arg0" type="xs:int"/>
          <xs:element name="arg1" type="xs:int"/>
        </xs:sequence>
      </xs:complexType>
      <xs:complexType name="sumResponse">
        <xs:sequence>
          <xs:element name="return" type="xs:int"/>
        </xs:sequence>
      </xs:complexType>
    </xs:schema>
  </wsdl:types>
  <wsdl:message name="sumResponse">
    <wsdl:part element="tns:sumResponse" name="parameters"/>
  </wsdl:message>
  <wsdl:message name="sum">
    <wsdl:part element="tns:sum" name="parameters"/>
  </wsdl:message>
  <wsdl:portType name="Sum">
    <wsdl:operation name="sum">
      <wsdl:input message="tns:sum" name="sum"/>
      <wsdl:output message="tns:sumResponse" name="sumResponse"/>
    </wsdl:operation>
  </wsdl:portType>
</wsdl:definitions>
```

Ingredients

Java SE Development Kit 21

<https://www.oracle.com/it/java/technologies/downloads/#java21>

I was using Java 1.8

Eclipse IDE for Enterprise Java Developers

<https://www.eclipse.org/downloads/packages>

I was using Eclipse 2023 - 03

Apache Tomcat® (v.11.0.5) [core - zip or tar.gz]

<https://tomcat.apache.org/download-11.cgi>

(see also <https://tomcat.apache.org/whichversion.html>)

I was using Apache Tomcat 9.0.73

Apache CXF™ (v.4.1.1) [binary distribution - zip or tar.gz]

<http://cxf.apache.org/download.html>

I was using Apache CXF 3.5.5
and version 4.0.3

SoapUI Open Source

<https://www.soapui.org/downloads/soapui.html>

Tomcat and CXF



Apache Tomcat®



Search...

COMMUNITY
THE ASF CONFERENCE
CODE

Apache Tomcat

Home
Toolbox

Tomcat 11 Software Downloads

Welcome to the Apache Tomcat® 11.x software download page. This page provides download links for obtaining the latest version of Tomcat 11.0.x software, as well as links to the archives of older releases.

Unsure which version you need? Specification versions implemented, minimum Java version required and lots more useful information may be found on the ['which version?'](#) page.

Users of Tomcat 11 onwards should be aware that, as a result of the move from Java EE to Jakarta EE as part of the transfer of Java EE to the Eclipse Foundation, the primary package for all implemented APIs has changed from javax.* to jakarta.*. This will almost certainly require code changes to enable applications to migrate from Tomcat 9 and earlier to Tomcat 11 and later. A [migration tool](#) has been developed to aid this process.

Links

Docs Home
FAQ

User Guide

- 1) Introduction
- 2) Setup
- 3) First webapp
- 4) Deployer
- 5) Manager
- 6) Host Manager
- 7) Realms and AAA

Documentation Index

Introduction

This is the top-level entry point of the documentation bundle for the **Apache Tomcat** Servlet/JSP container. Apache Tomcat version 11.0 implements the Servlet 6.0 and Pages 4.0 [specifications](#) from [Jakarta EE](#), and includes many additional features that make it a useful platform for developing and deploying web applications and web services.

Select one of the links from the navigation menu (to the left) to drill down to the more detailed documentation that is available. Each available manual is described in more detail below.

Tomcat and CXF

Apache CXF

[Index](#) > [Download](#)

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APACHE CXF

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- [People](#)
- [Project Status](#)
- [Roadmap](#)
- [Mailing Lists](#)
- [Issue Reporting](#)
- [Special Thanks](#)
- [License](#)
- [Security Advisories](#)

USERS

- [User's Guide](#)
- [Support](#)
- [FAQ](#)
- [Resources and Articles](#)

SEARCH

Releases

4.1.1

The 4.1.1 is our release to feature Jakarta EE 10 support and JDK-17 baseline. For a complete list of new features, API changes, etc... please see the [release notes](#) and [migration guide](#).

Description	File	sha256	PGP
Source distribution	apache-cxf-4.1.1-src.tar.gz	apache-cxf-4.1.1-src.tar.gz.sha256	
	apache-cxf-4.1.1-src.zip	apache-cxf-4.1.1-src.zip.sha256	
Binary distribution	apache-cxf-4.1.1.tar.gz	apache-cxf-4.1.1.tar.gz.sha256	
	apache-cxf-4.1.1.zip	apache-cxf-4.1.1.zip.sha256	

Installation Prerequisites

Before installing Apache CXF, make sure the following products, with the specified versions, are installed on your system:

- Java 17 Development Kit
- Maven 3.9 or later

Java Development Kit

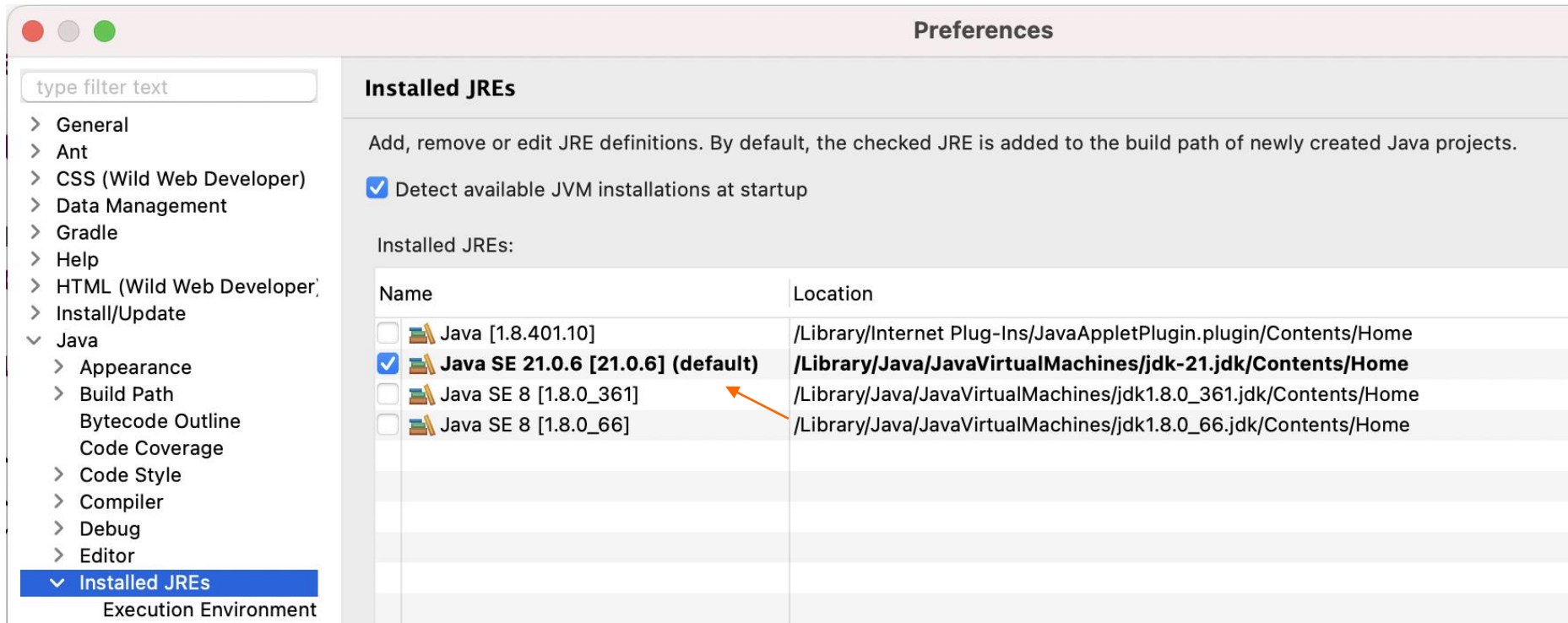
You must install the J2SE Development Kit (JDK) 17.0, which can be downloaded from:

<http://www.oracle.com/technetwork/java/javase/downloads/index.html>

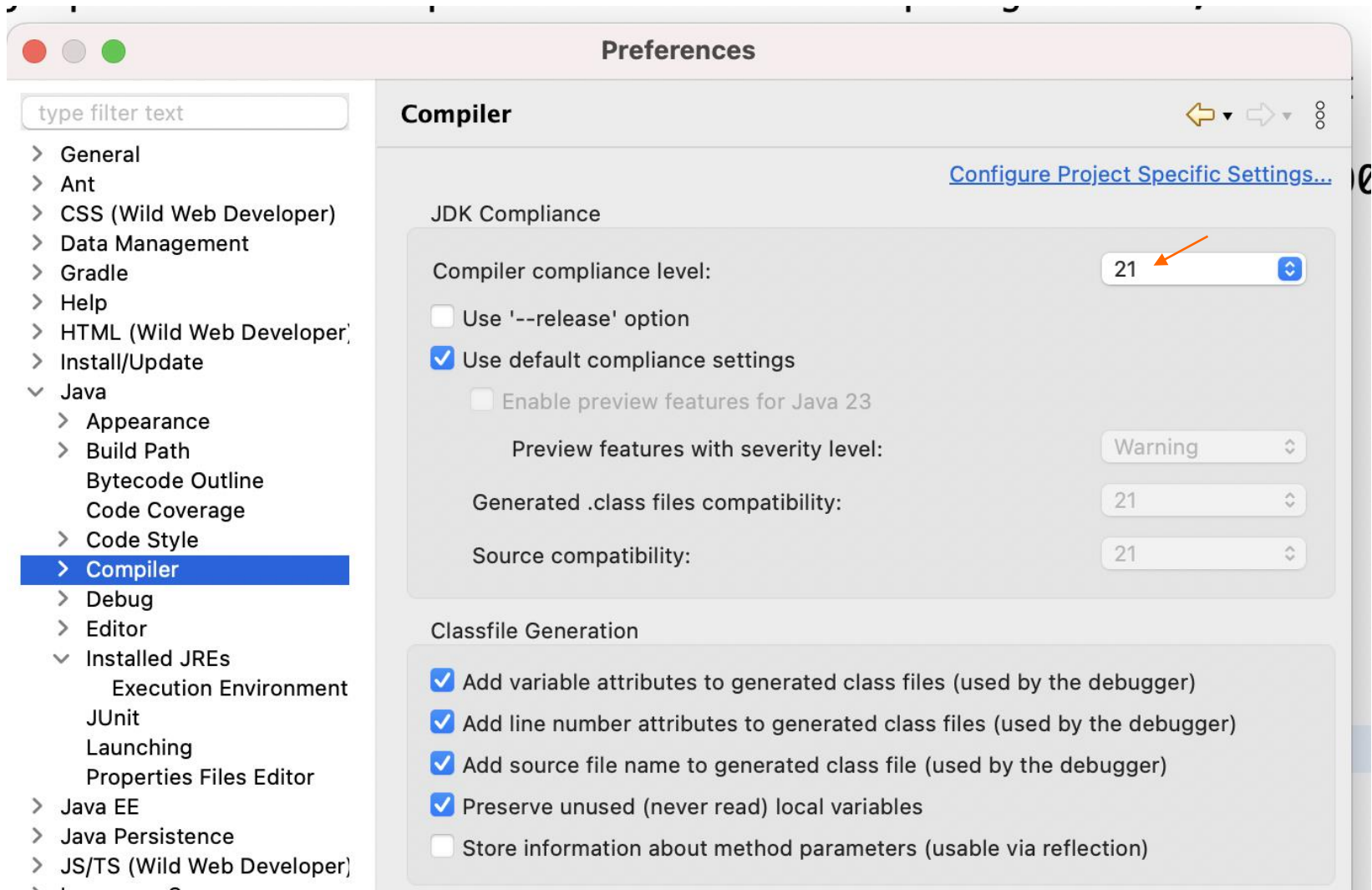
After installing the JDK, set or modify the following environment variables:

- JAVA_HOME – set this environment variable to point at the root directory of the Java installation.
- PATH – make sure that your PATH includes:
%JAVA_HOME%\bin (Windows)
\$JAVA_HOME/bin (UNIX)

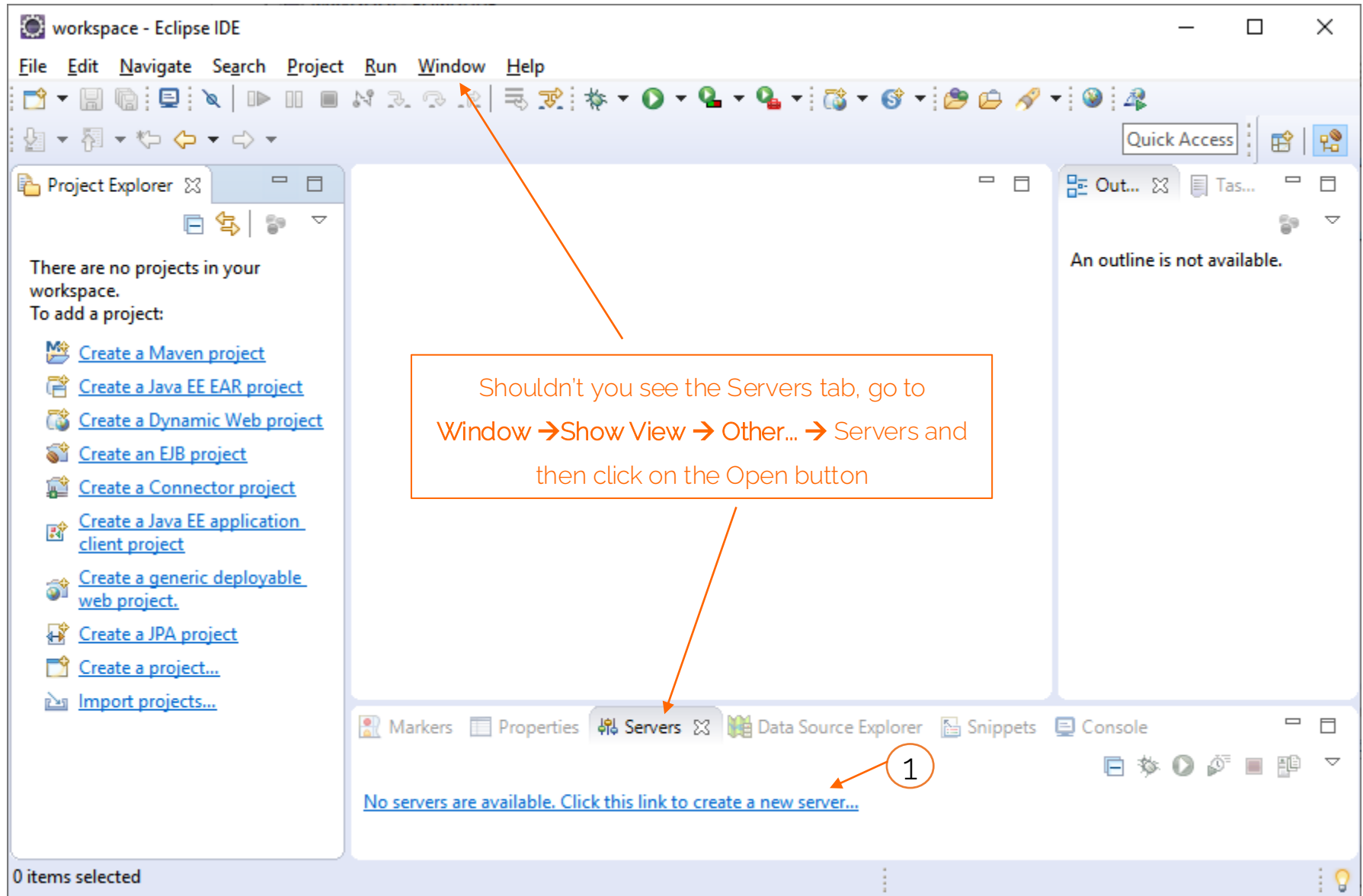
As usual, be sure that the right JDK is selected...



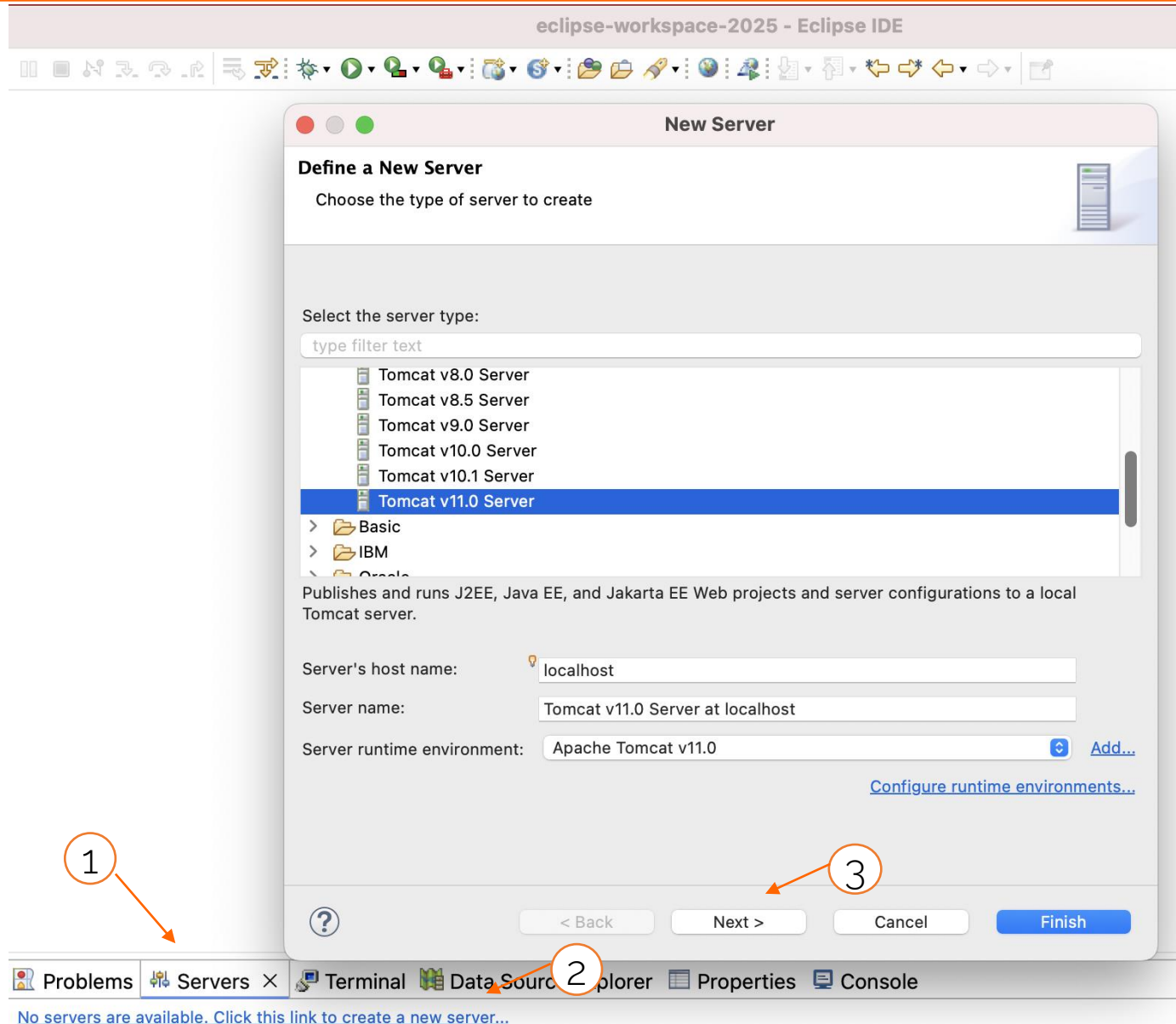
... and the compiler compliance level as well



Eclipse IDE for Enterprise Java Developers



Configure Tomcat



Hint on managing runtime environments

The screenshot shows the Eclipse IDE interface. On the left, the 'Eclipse' menu is open, with 'Settings...' selected. Below this, a list of preferences categories is shown, with 'Runtime Environments' highlighted. An orange box with the text 'Global preferences' is overlaid on the left side. The main window displays the 'Preferences' dialog, specifically the 'Server Runtime Environments' tab. This tab contains a table with two columns: 'Name' and 'Type'. The table lists two environments: 'Apache Tomcat v11.0' and 'TomEE v10.0.0', both of type 'Apache Tomcat v11.0'. An orange arrow points to the text 'Add, remove, or edit server runtime environments.' above the table. The bottom of the dialog has 'Cancel' and 'Apply and Close' buttons.

Global preferences

Preferences

Server Runtime Environments

Add, remove, or edit server runtime environments.

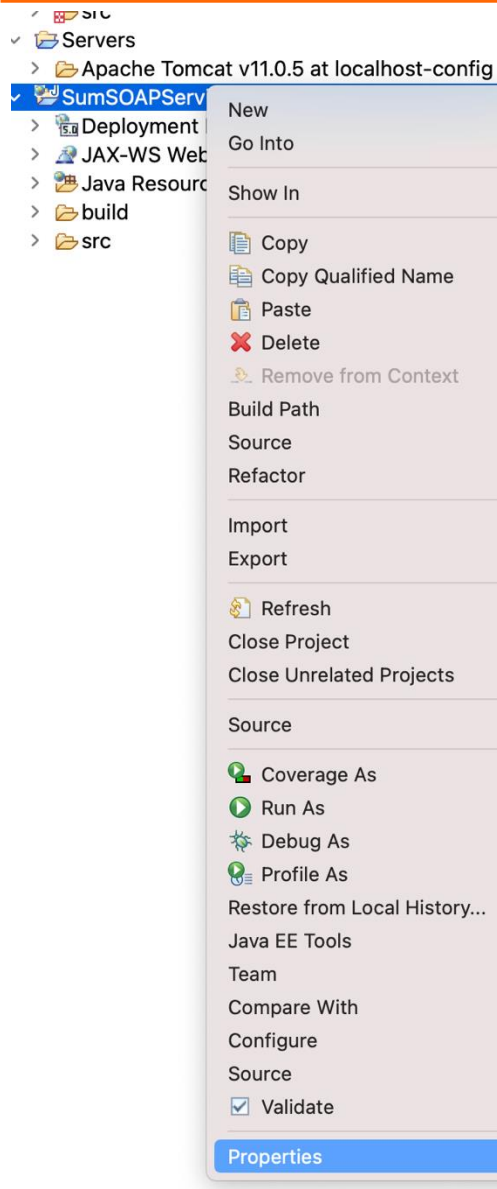
Server runtime environments:

Name	Type
Apache Tomcat v11.0	Apache Tomcat v11.0
TomEE v10.0.0	Apache Tomcat v10.1

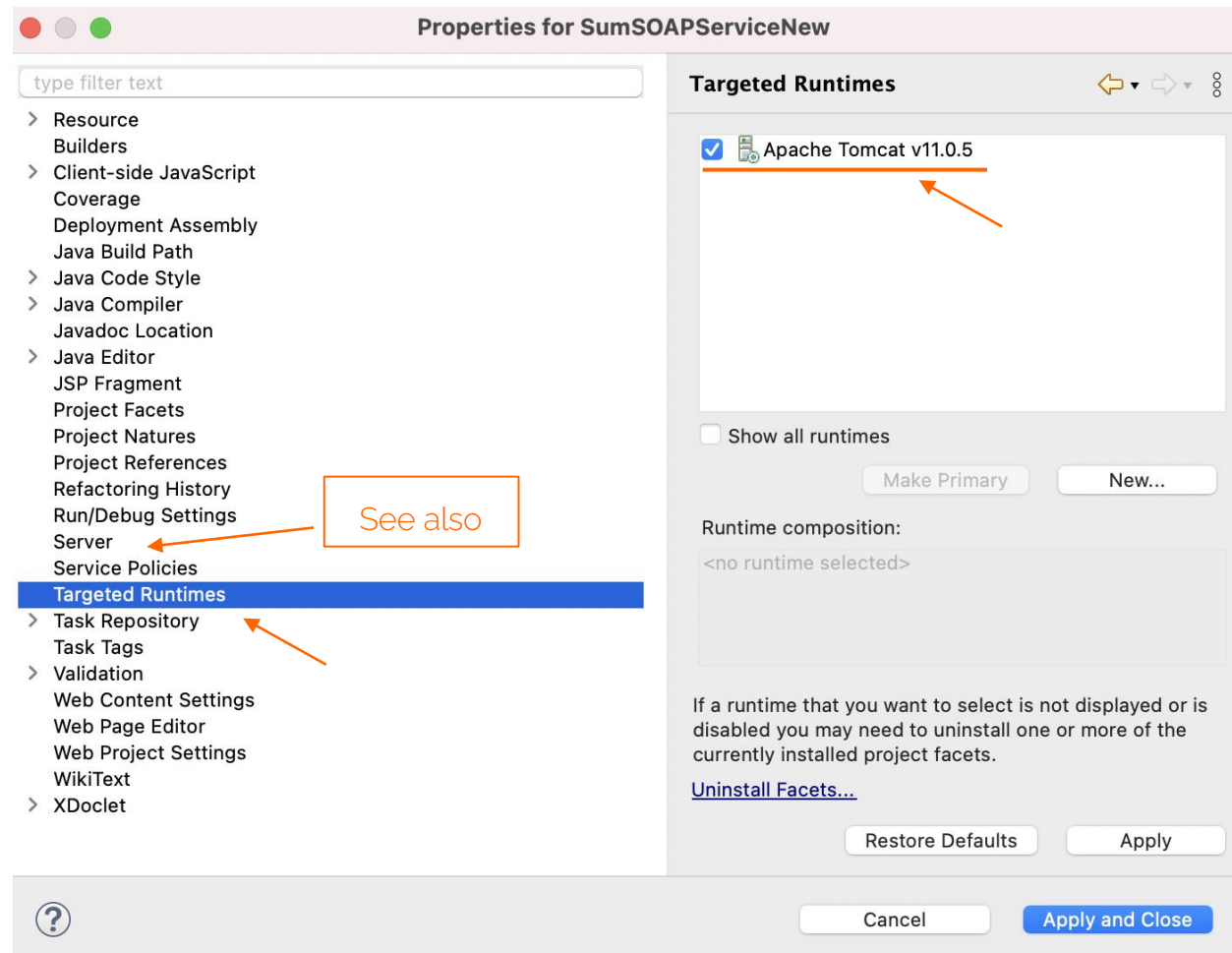
Buttons: Add..., Edit..., Remove, Search..., Columns...

Buttons: Cancel, Apply and Close

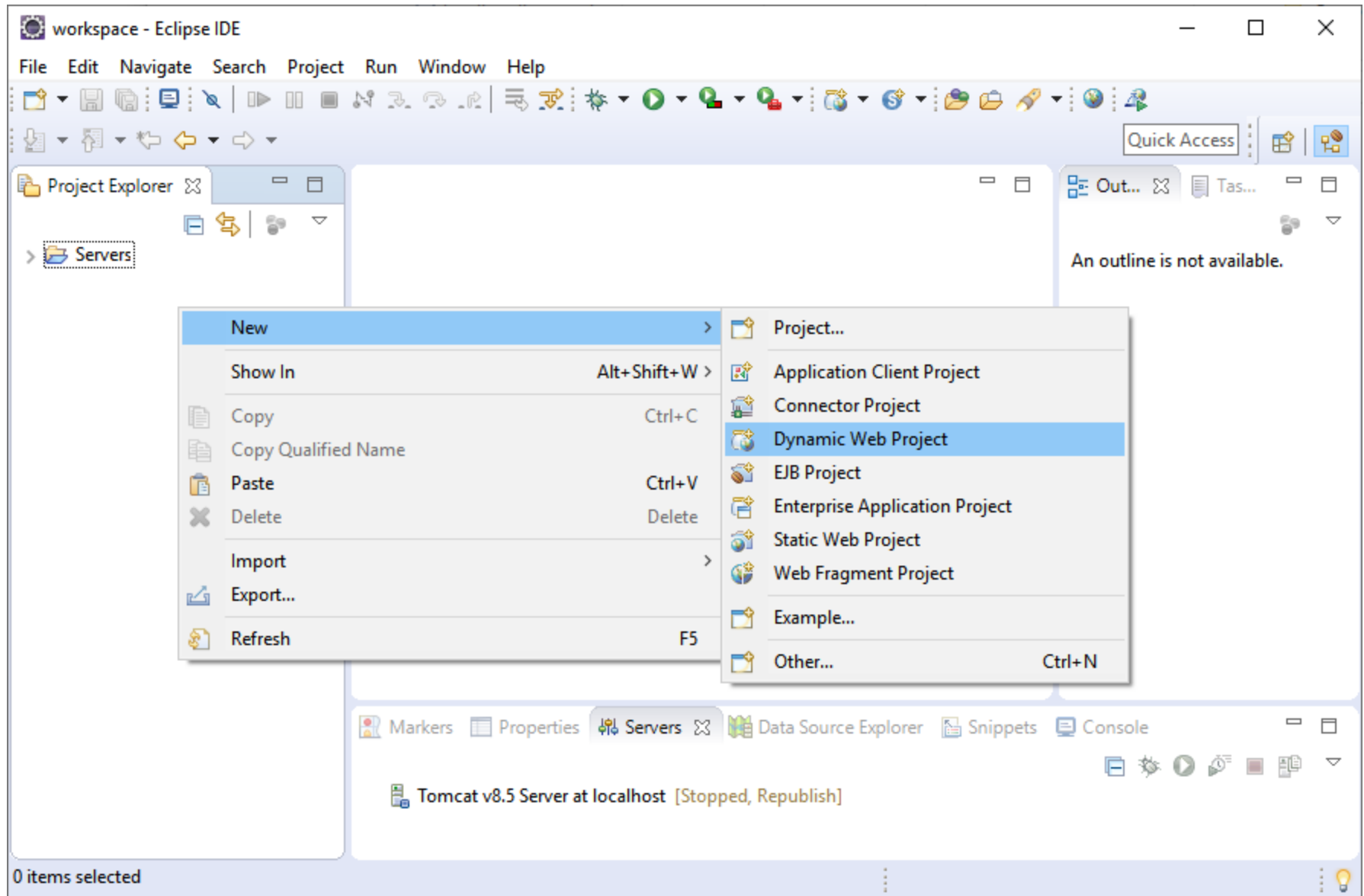
Hint on managing runtime environments



Project-level
properties



New Dynamic Web Project



New Dynamic Web Project

Remember that the **Dynamic Web Module** version correlates with the Servlet API version.

We will see our first servlet on the next slide

A Servlet is an object that receives a request and sends a response back based on that request.

New Dynamic Web Project

Create a standalone Java-based Web Application or add it to a new or existing Enterprise Application.

Project name: **SumSOAPServiceNew** (1)

Project location

☒ Use default location

Location: /Users/marcoautili/Library/CloudStorage/OneDrive-Universita'degliStudidell'Aquila/TEACHING/eclipse-wo (2) Browse...

Target runtime

Apache Tomcat v11.0 (3) New Runtime...

Dynamic web module version

6.0

Configuration

<custom> Modify...

Hint: Get started quickly by selecting one of the pre-defined project configurations.

EAR membership

☐ Add project to an EAR

EAR project name: SumSOAPServiceNeEAR New Project...

Working sets

☐ Add project to working sets New...

Working sets: Select...

4 DO NOT CLICK on Finish 😊

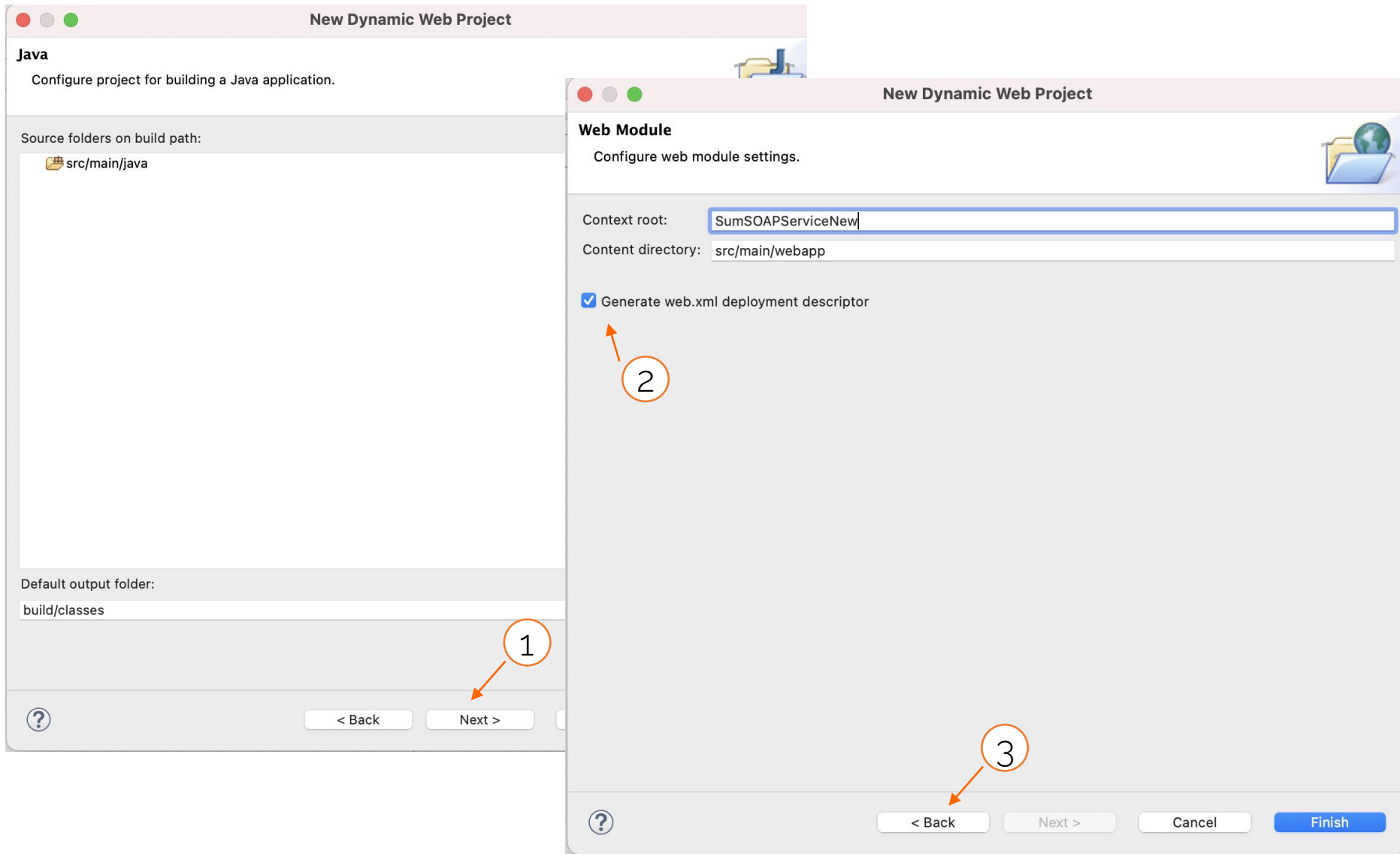
< Back Next > Cancel Finish

Servlet API history

Servlet API version	Released	Specification	Platform	Important Changes
Jakarta Servlet 6.0	May 31, 2022	6.0	Jakarta EE 10	remove deprecated features and implement requested enhancements
Jakarta Servlet 5.0	Oct 9, 2020	5.0	Jakarta EE 9	API moved from package <code>javax.servlet</code> to <code>jakarta.servlet</code>
Jakarta Servlet 4.0.3	Sep 10, 2019	4.0	Jakarta EE 8	Renamed from "Java" trademark
Java Servlet 4.0	Sep 2017	JSR 369	Java EE 8	HTTP/2
Java Servlet 3.1	May 2013	JSR 340	Java EE 7	Non-blocking I/O, HTTP protocol upgrade mechanism (WebSocket) [19]
Java Servlet 3.0	December 2009	JSR 315	Java EE 6, Java SE 6	Pluggability, Ease of development, Async Servlet, Security, File Uploading
Java Servlet 2.5	September 2005	JSR 154	Java EE 5, Java SE 5	Requires Java SE 5, supports annotation

At least Jakarta EE 10 is required

New Dynamic Web Project



Put the CXF jars in the lib folder (not all jars)

SumSOAPServiceNew

- Deployment Descriptor: SumSOAPServiceNew
- JAX-WS Web Services
- Java Resources
- build
- src
 - main
 - java
 - webapp
 - META-INF
 - WEB-INF
 - lib
 - asm-9.7.1.jar
 - cxf-core-4.1.1.jar
 - cxf-rt-bindings-soap-4.1.1.jar
 - cxf-rt-bindings-xml-4.1.1.jar
 - cxf-rt-databinding-jaxb-4.1.1.jar
 - cxf-rt-frontend-jaxws-4.1.1.jar
 - cxf-rt-frontend-simple-4.1.1.jar
 - cxf-rt-transport-http-4.1.1.jar
 - cxf-rt-wsdl-4.1.1.jar
 - istack-commons-runtime-4.1.2.jar
 - jakarta.activation-api-2.1.3.jar
 - jakarta.servlet-api-6.0.0.jar
 - jakarta.xml.bind-api-4.0.2.jar
 - jakarta.xml.soap-api-3.0.2.jar
 - jakarta.xml.ws-api-4.0.2.jar
 - jaxb-core-4.0.5.jar
 - jaxb-runtime-4.0.5.jar
 - stax2-api-4.2.2.jar
 - txw2-4.0.5.jar
 - woodstox-core-7.1.0.jar
 - wsdl4j-1.6.3.jar
 - xmlschema-core-2.3.1.jar

META-INF

WEB-INF

lib

web.xml

asm-9.1.jar

cxf-core-3.4.3.jar

cxf-rt-bindings-soap-3.4.3.jar

cxf-rt-databinding-jaxb-3.4.3.jar

cxf-rt-frontend-jaxws-3.4.3.jar

cxf-rt-frontend-simple-3.4.3.jar

cxf-rt-transport-http-3.4.3.jar

cxf-rt-wsdl-3.4.3.jar

javax.servlet-api-3.1.0.jar

stax2-api-4.2.1.jar

woodstox-core-6.2.4.jar

wsdl4j-1.6.3.jar

xmlschema-core-2.2.5.jar

These are the jars that I had to add for the older version «apache-cxf-3.4.»

Copy these jars from the folder «apache-cxf-4.1.1/lib»

How did I select these specific jars among ~240?

- Is this the way to go?
- Is this the best we can do?

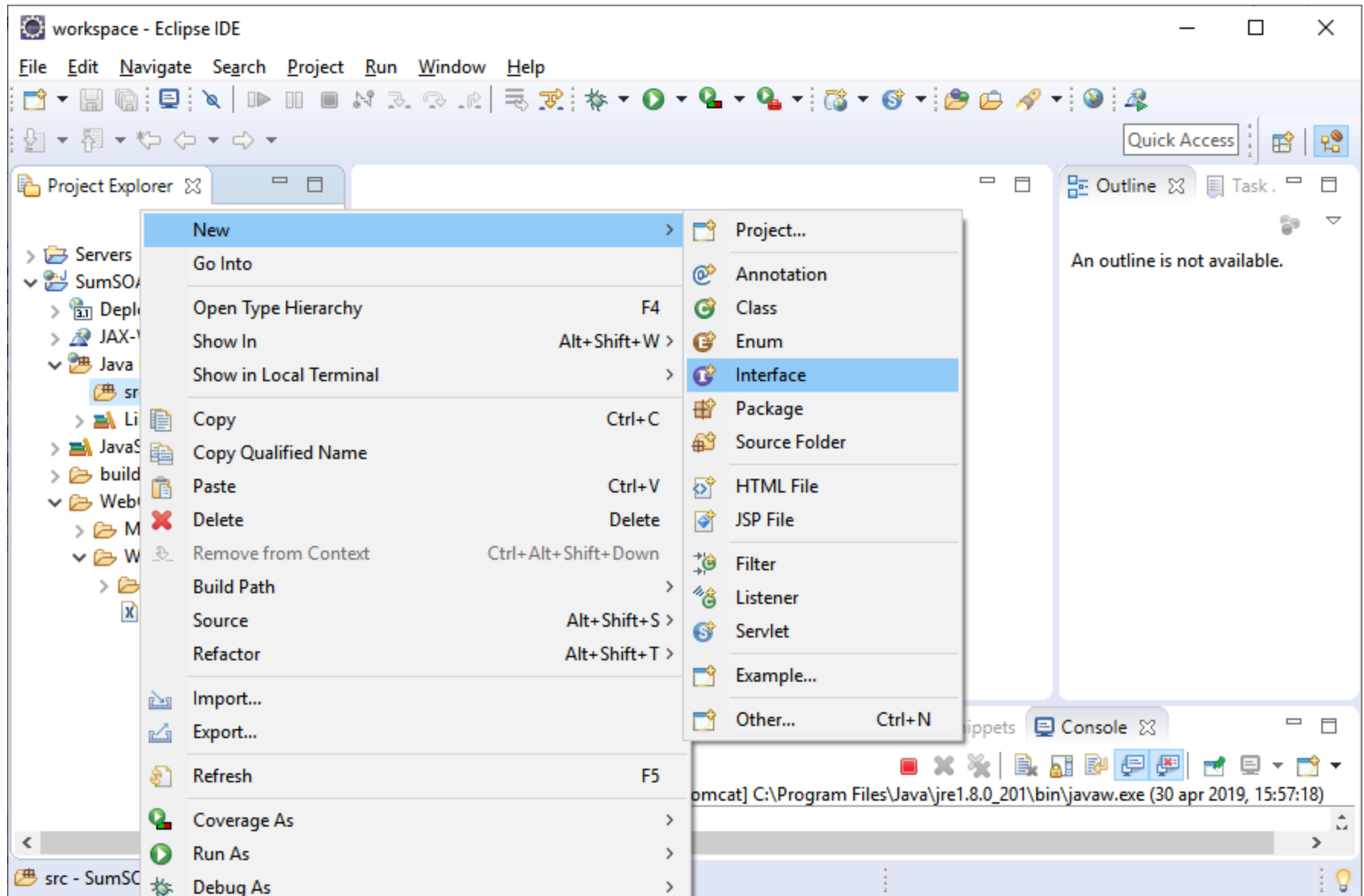
We'll see... 😊

Updated example

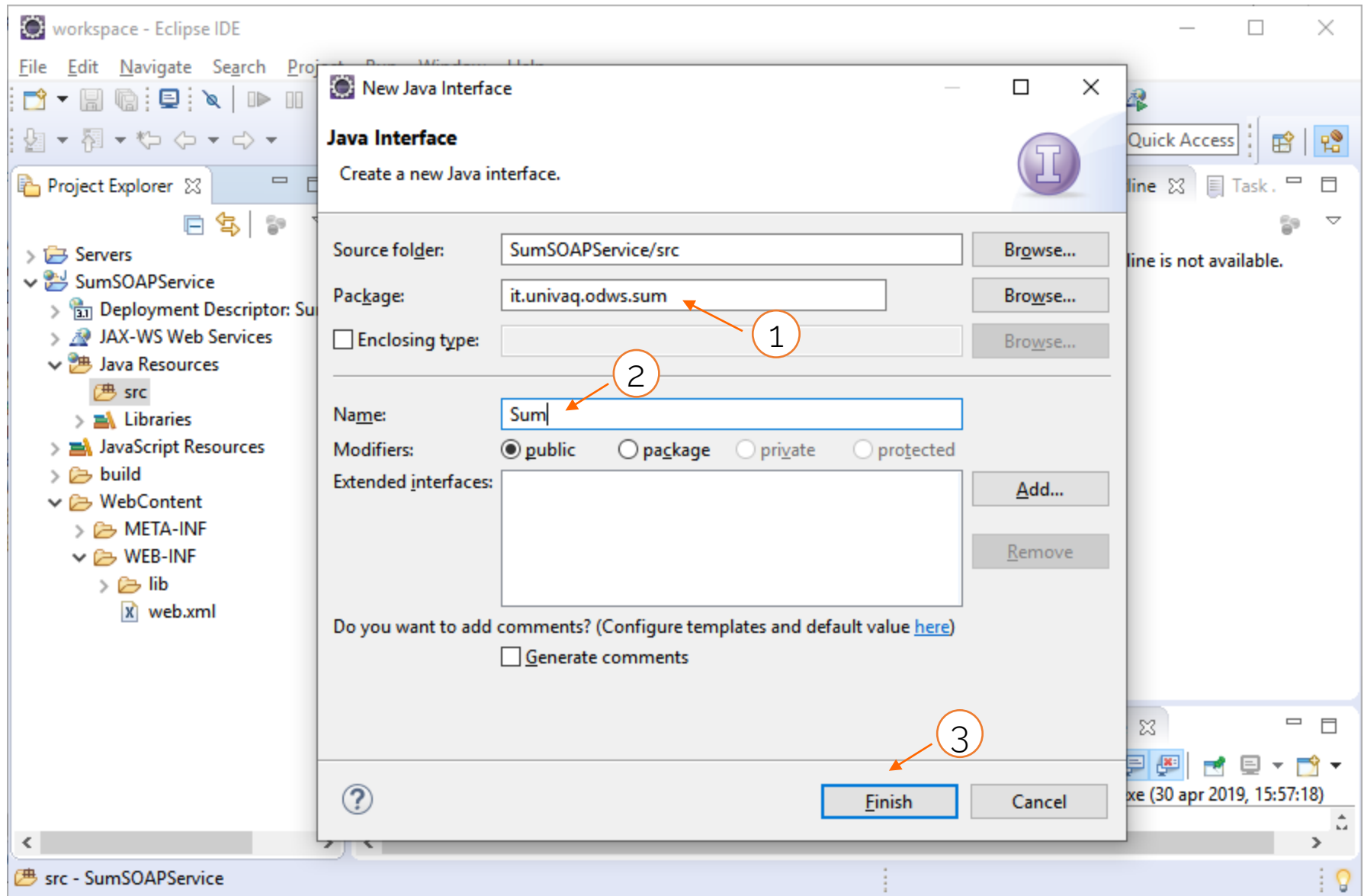
From Java 8 to Java 21 (Jakarta EE 10) with newer Tomcat and CXF versions

- I did not fully update the following slides with the new screenshots/versions/text... the "migration" is now straightforward 😊
- Some of the notable changes to be accounted for:
 - Project renaming: `SumSOAPService` → `SumSOAPServiceNew`
 - Package renaming: `javax` → `jakarta`
 - `Number` and `version` of included `jars` from `apache-cxf-4.1.1/lib`
 - ...

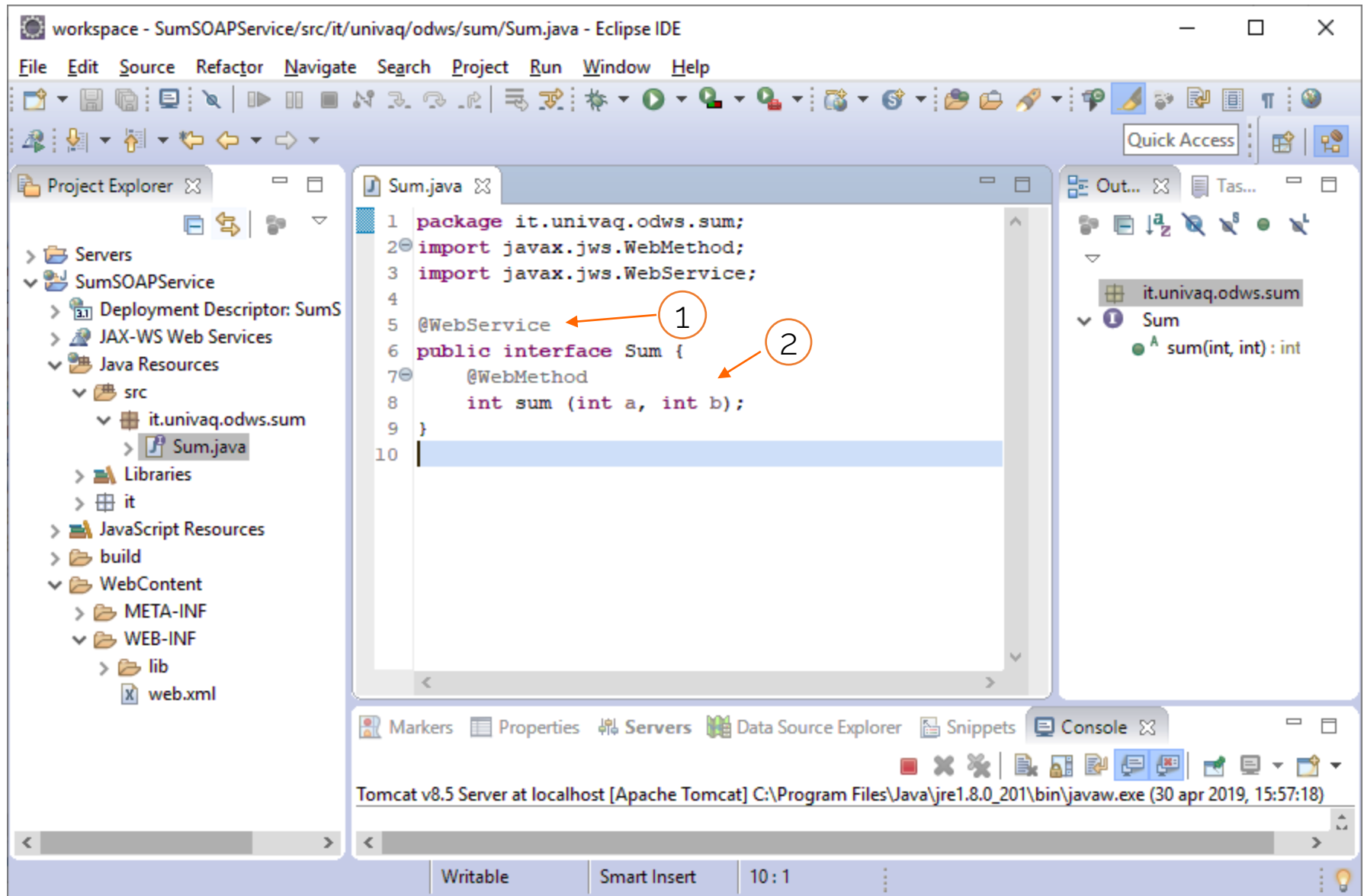
Create a service interface



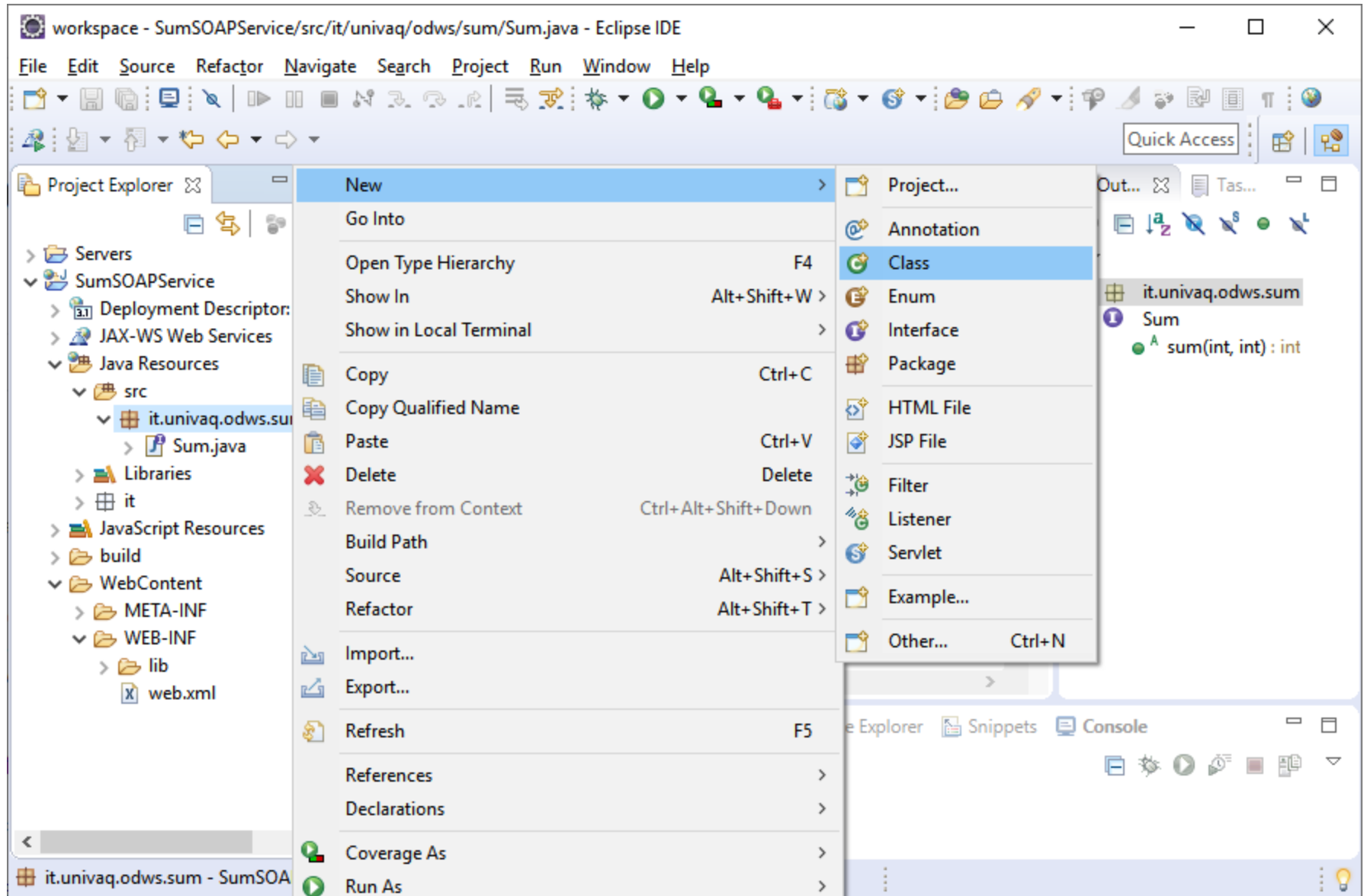
Create a service interface



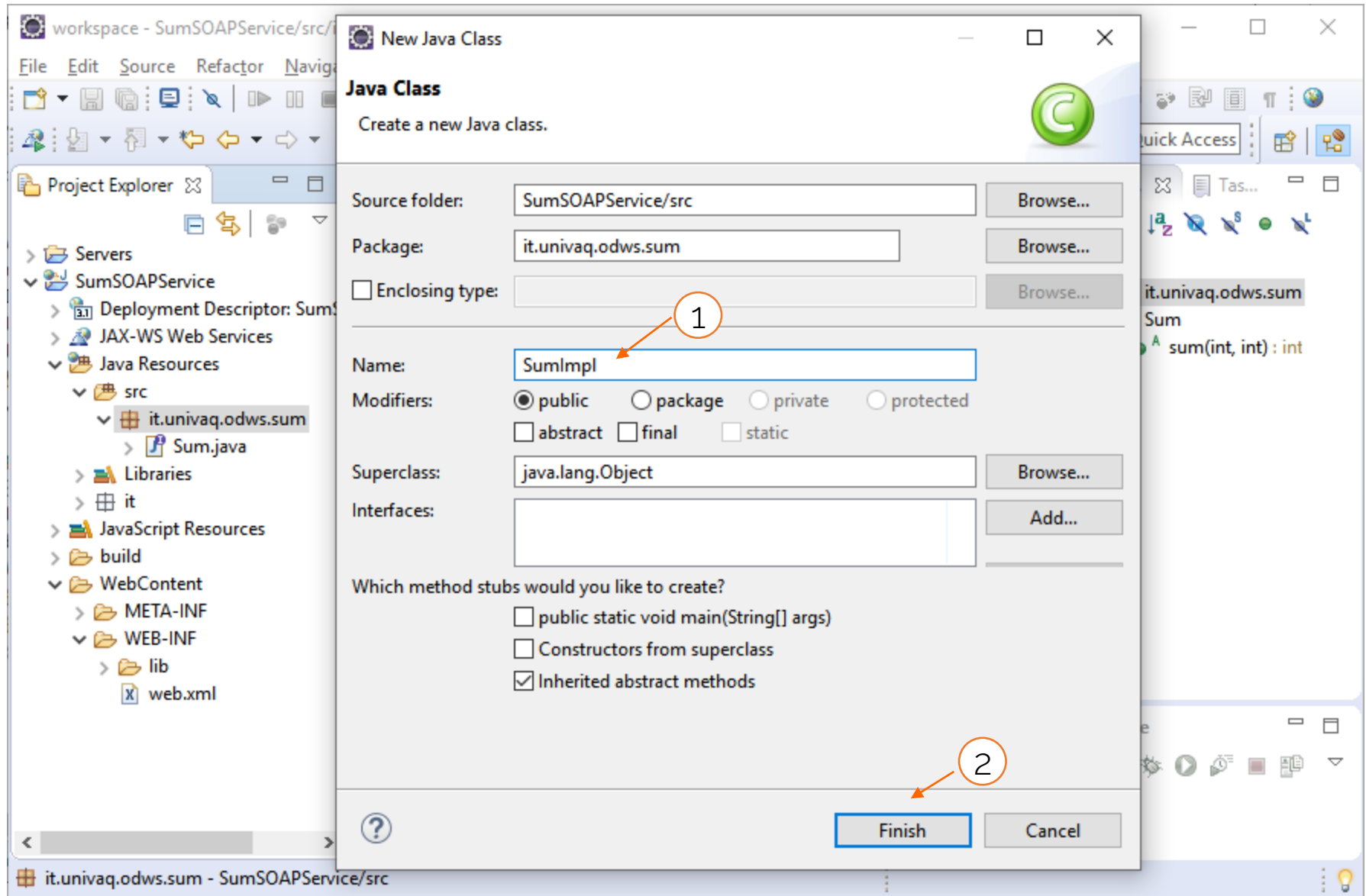
Create a service interface



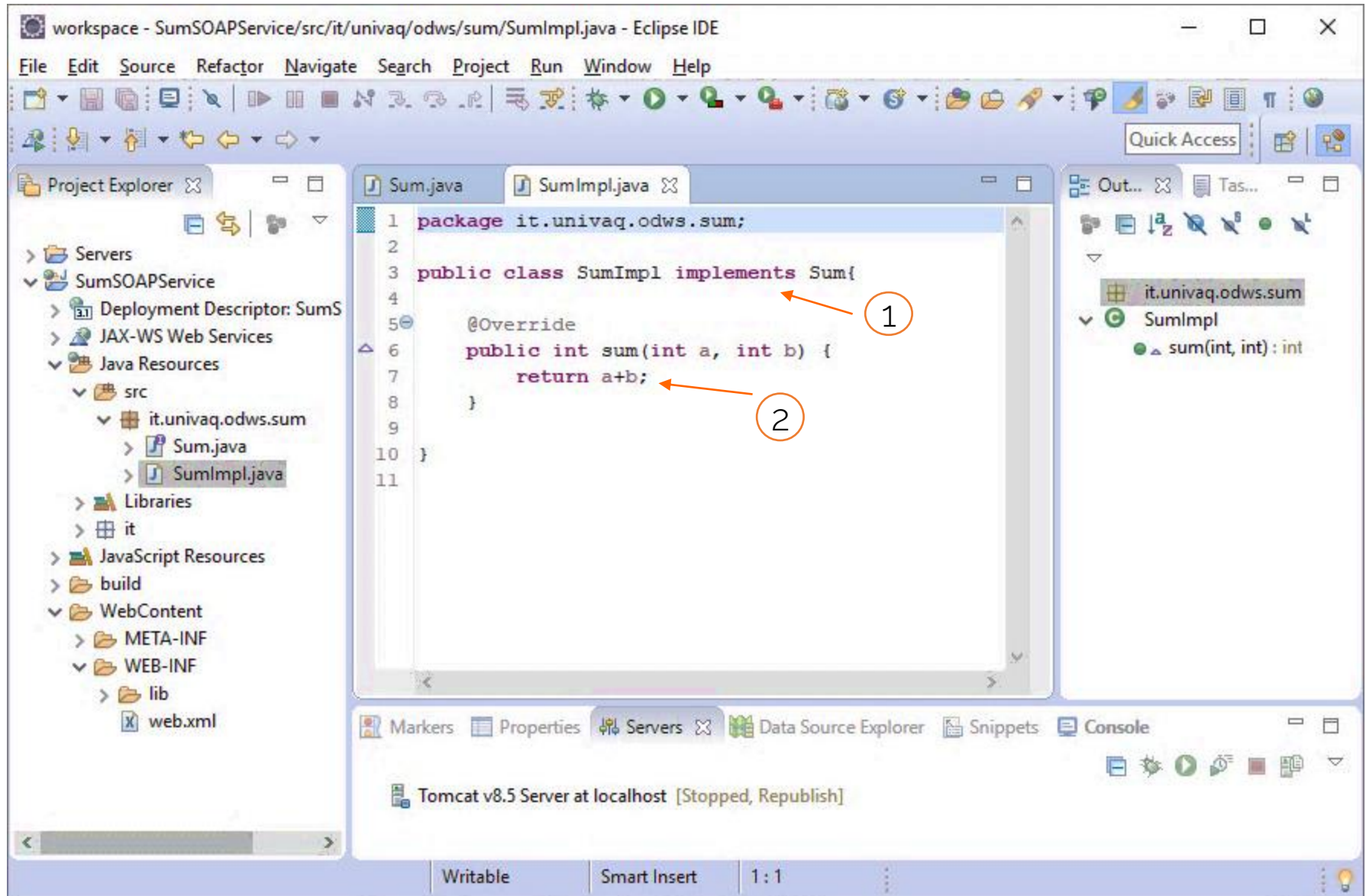
Implement the service interface



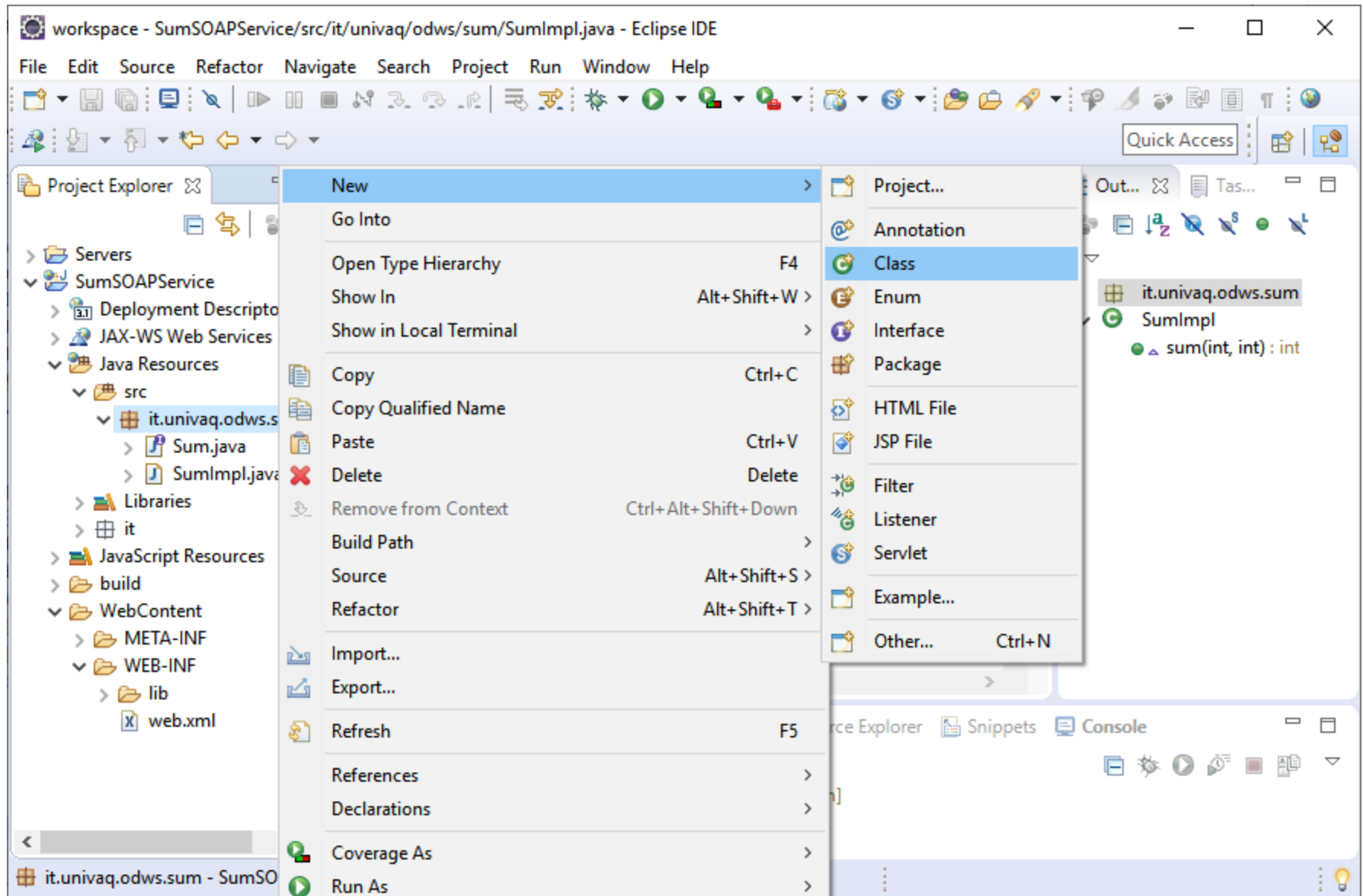
Implement the service interface



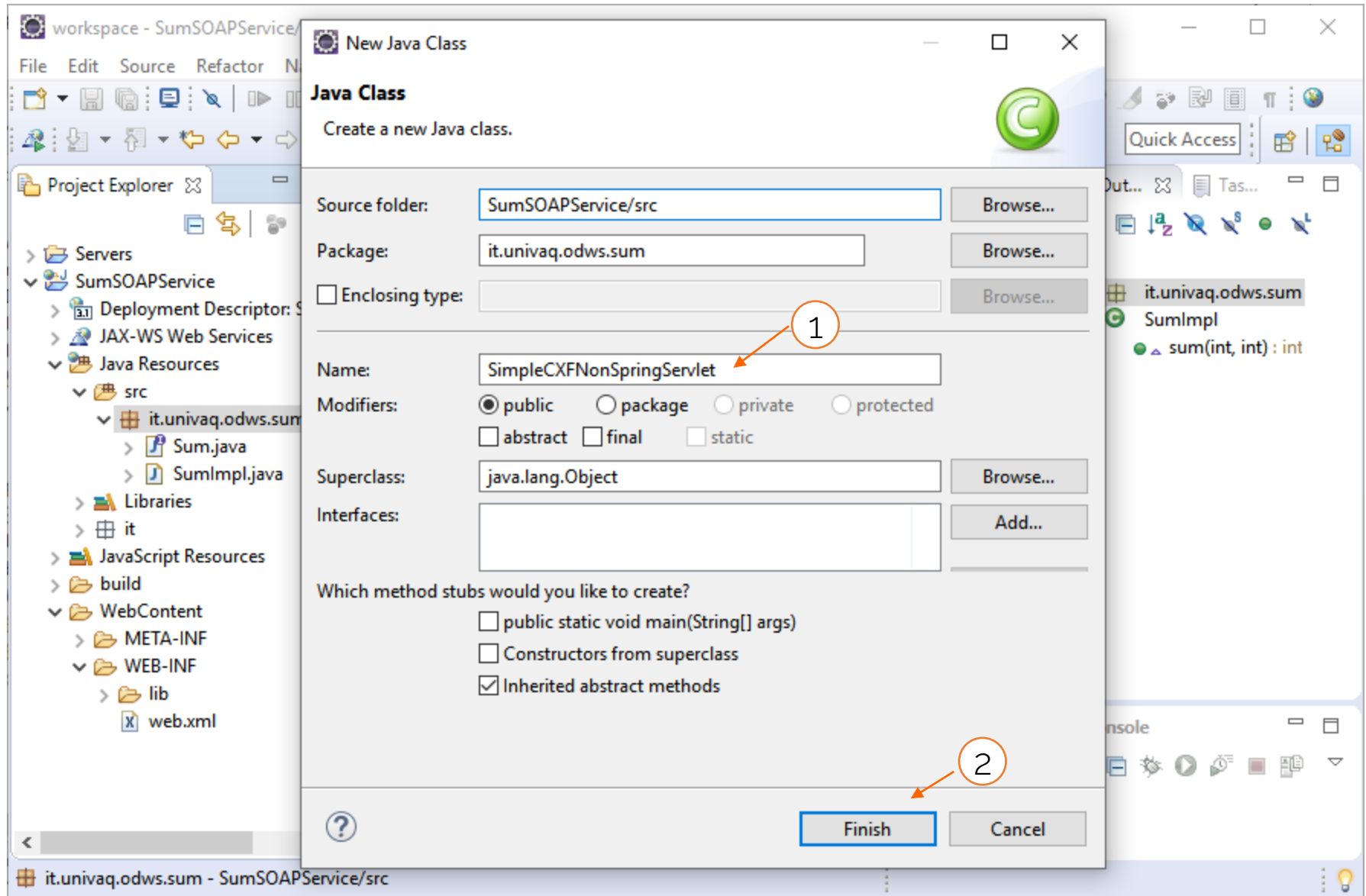
Implement the service interface



Create a simple servlet



Create a simple servlet



Create a CXFNonSpringServlet

Using the servlet transport without Spring

A user who doesn't want to touch any Spring stuff could also publish the endpoint with CXF servlet transport. First you should **extend the CXFNonSpringServlet** and then **override the method loadBus**, e.g.:

```
import javax.xml.ws.Endpoint;
...

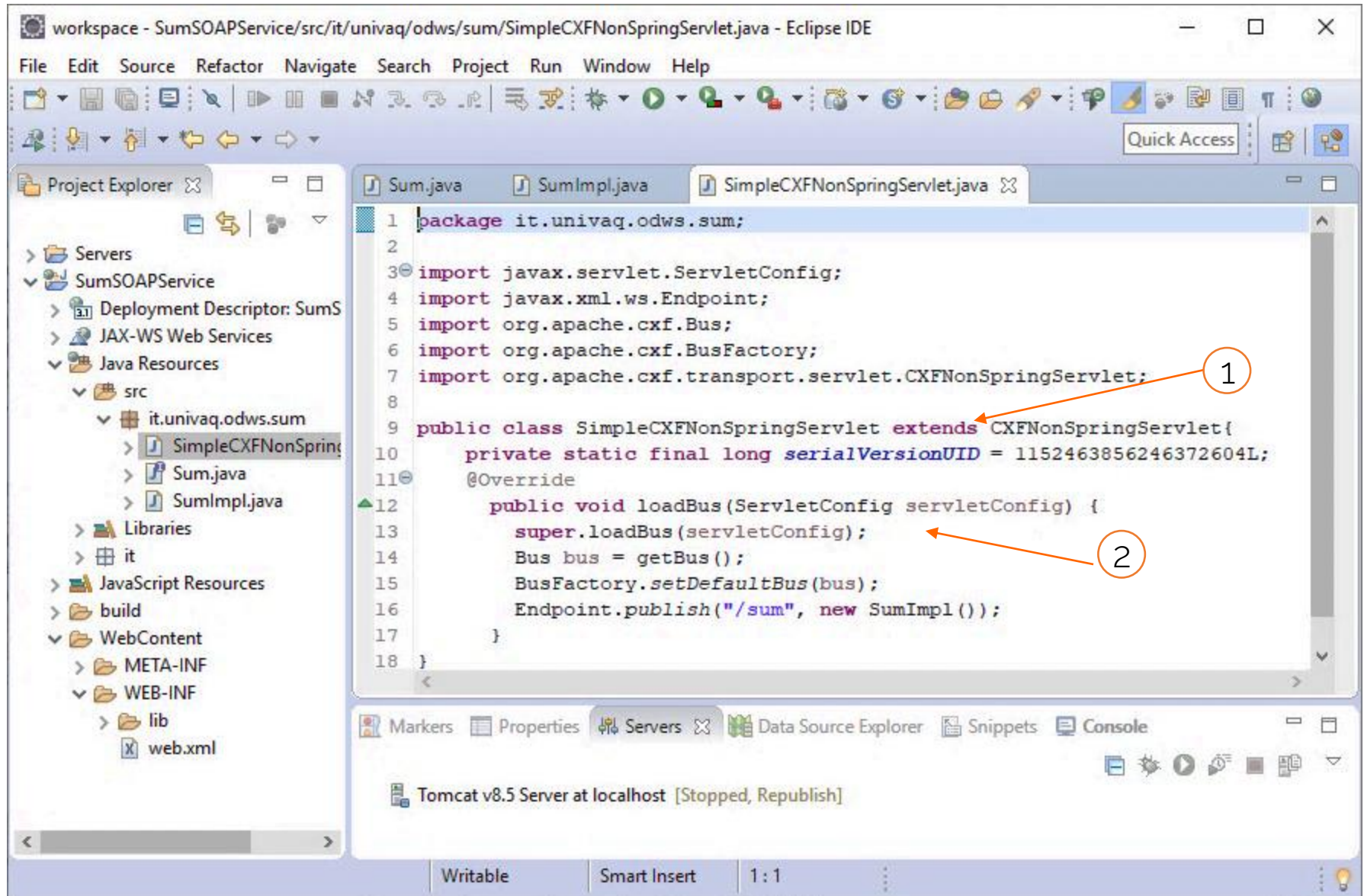
@Override
public void loadBus(ServletConfig servletConfig) throws ServletException {
    super.loadBus(servletConfig);

    // You could add the endpoint publish codes here
    Bus bus = cxf.getBus();
    BusFactory.setDefaultBus(bus);
    Endpoint.publish("/Greeter", new GreeterImpl());

    // You can also use the simple frontend API to do this
    ServerFactoryBean factory = new ServerFactoryBean();
    factory.setBus(bus);
    factory.setServiceClass(GreeterImpl.class);
    factory.setAddress("/Greeter");
    factory.create();
}
```

<https://cxf.apache.org/docs/servlet-transport.html>

Create a CXFNonSpringServlet



Configure the servlet in the web.xml file

The screenshot shows the Eclipse IDE interface with the following components:

- Project Explorer:** Displays the project structure for 'SumSOAPService'. The 'WEB-INF' directory is expanded, showing 'lib' and 'web.xml'.
- Editor:** Shows the 'web.xml' file with the following XML content:

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://xmlns.jcp.org/xml/ns/javaee" xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee http://xmlns.jcp.org/xml/ns/javaee/web-app_3_0.xsd">
  <display-name>SumSOAPService</display-name>
  <servlet>
    <servlet-name>SimpleCXFNonSpringServlet</servlet-name>
    <servlet-class>it.univaq.odws.sum.SimpleCXFNonSpringServlet</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>SimpleCXFNonSpringServlet</servlet-name>
    <url-pattern>/*</url-pattern>
  </servlet-mapping>
  <welcome-file-list>
    <welcome-file>index.html</welcome-file>
    <welcome-file>index.htm</welcome-file>
    <welcome-file>index.jsp</welcome-file>
  </welcome-file-list>
</web-app>
```

Annotations in the image:

 - A red arrow labeled '1' points to the `<servlet>` tag on line 5.
 - A red arrow labeled '2' points to the `<servlet-mapping>` tag on line 9.
 - A red box highlights the URL patterns in the `<servlet-mapping>` section, showing the mapping from `http://localhost:8080/SumSOAPService` to `http://localhost:8080/SumSOAPServiceNew /`.
- Console:** Shows the output of the Tomcat v8.5 Server at localhost, indicating the server is running and the application is initialized.

Eclipse → Settings

The screenshot shows the Eclipse IDE interface. On the left, the 'web.xml' file is open, displaying XML code. Line 2 has an error icon (a red 'x') and a tooltip that reads: 'Multiple annotations found at this line: - There is '1' error in 'jakartaee_10.xsd'. - Downloading external resources is disabled.' An orange arrow points from a text box at the bottom left to this error. The text box contains the text: 'Should you have this problem in the file web.xml ...'. On the right, the 'XML (Wild Web Developer)' settings page is open. It has a search bar at the top with the text 'type filter text'. Below it, the 'General' section is expanded. It contains the text 'See [XML Catalogs](#) for XML catalogs preferences' and two checkboxes: 'Download external resources like referenced DTD, XSD' (unchecked) and 'Enable autoclosing of XML tags' (checked). An orange arrow points from the 'Enable autoclosing of XML tags' checkbox to a text box that says '... Select'. At the bottom of the settings page, a list of categories is shown, with 'XML (Wild Web Developer)' selected and highlighted in blue. Other categories include Mylyn, Oomph, Plug-in Development, Run/Debug, Server, Terminal, TextMate, Validation, Version Control (Team), Web, Web Services, and XML.

Sum.java web.xml ×

```
1 <?xml version="1.0" encoding="UTF-8" standalone="no" ?>
2 <!-- Multiple annotations found at this line:
3      - There is '1' error in 'jakartaee_10.xsd'.
4      - Downloading external resources is disabled. -->
5 <display-name>SumSOAPService</display-name>
<welcome-file-list>
<welcome-file>index.html</welcome-file>
```

XML (Wild Web Developer)

See [XML Catalogs](#) for XML catalogs preferences

☐ Download external resources like referenced DTD, XSD

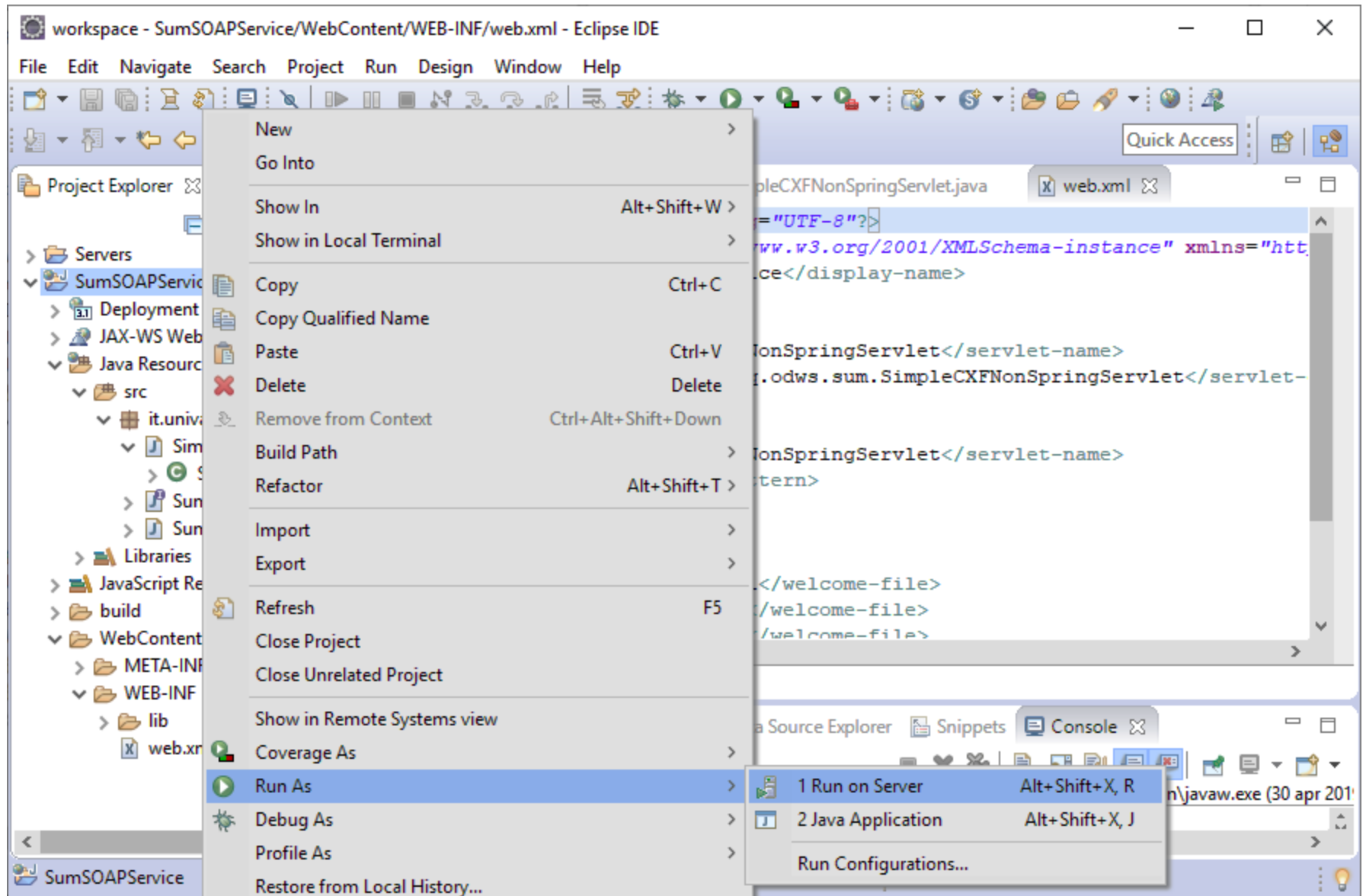
☒ Enable autoclosing of XML tags

... Select

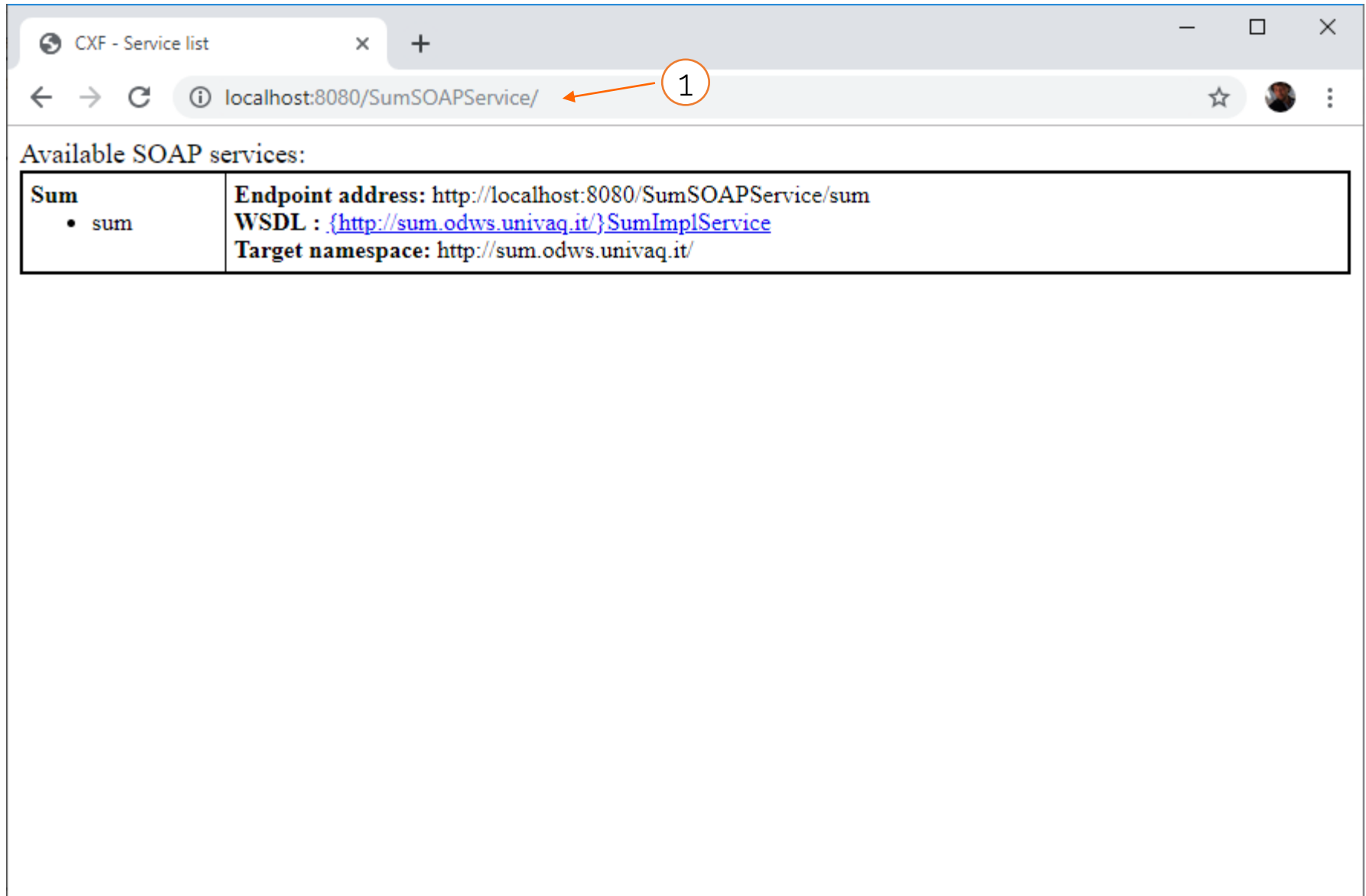
Mylyn
Oomph
Plug-in Development
Run/Debug
Server
Terminal
TextMate
Validation
Version Control (Team)
Web
Web Services
XML
XML (Wild Web Developer)
YAML (Wild Web Developer)

Should you have this problem in the file web.xml ...

Deploy and run the service on Tomcat



The service is up and running

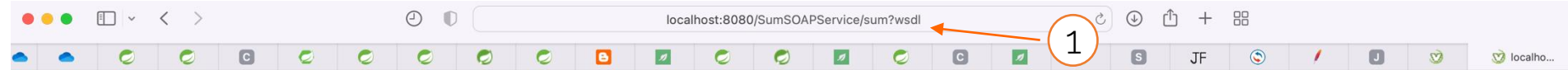


CXF - Service list

localhost:8080/SumSOAPService/ 1

Available SOAP services:

Sum sum	Endpoint address: http://localhost:8080/SumSOAPService/sum WSDL : http://sum.odws.univaq.it/SumImplService Target namespace: http://sum.odws.univaq.it/
--	---



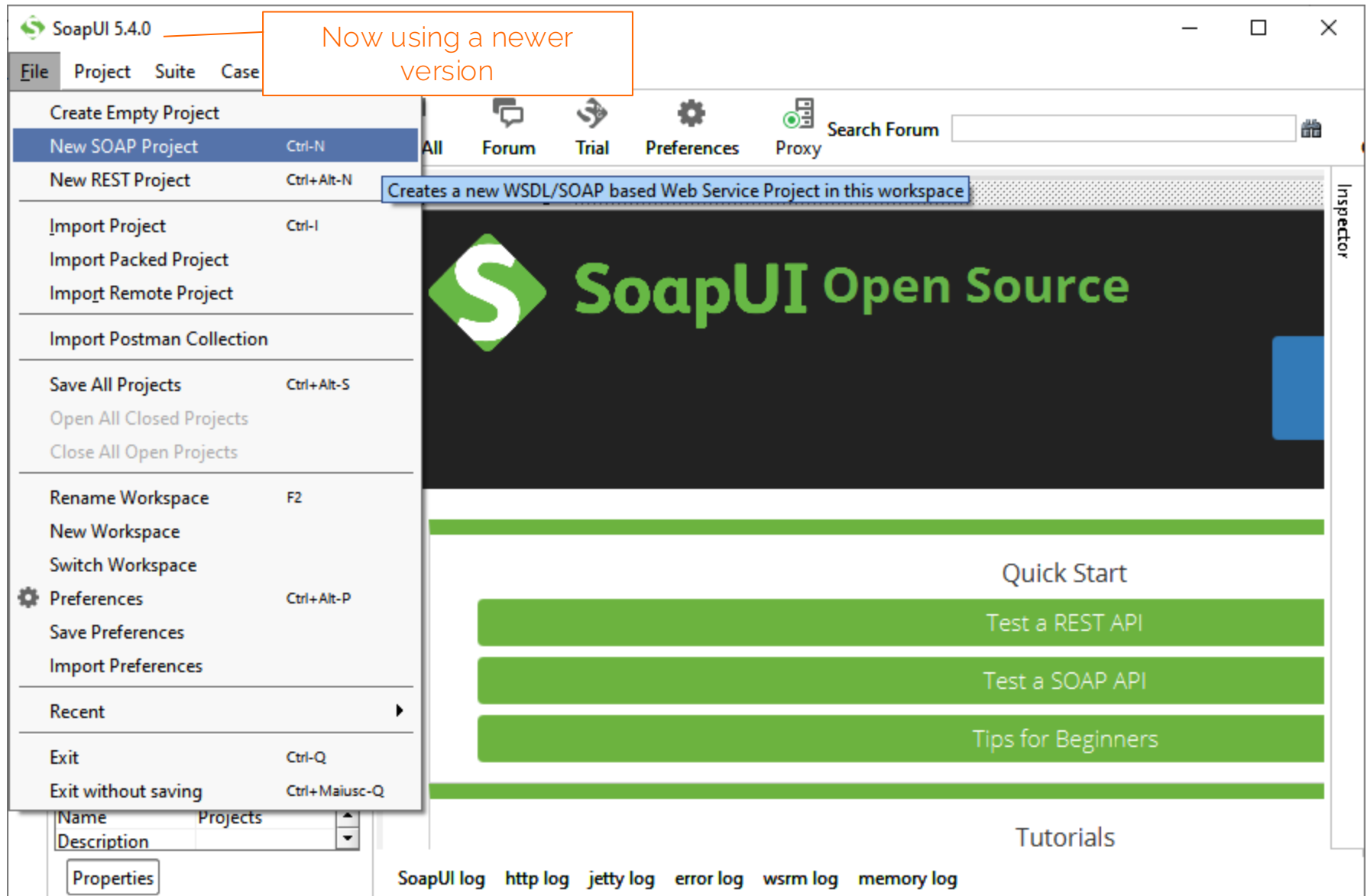
This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8" ?>
<wsdl:definitions xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:wsdl="http://schemas.xmlsoap.org/wsdl/"
  xmlns:tns="http://sum.odws.univaq.it/" xmlns:soap="http://schemas.xmlsoap.org/wsdl/soap/"
  xmlns:ns1="http://schemas.xmlsoap.org/soap/http" name="SumImplService" targetNamespace="http://sum.odws.univaq.it/">
  <wsdl:types>
    <xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:tns="http://sum.odws.univaq.it/" elementFormDefault="unqualified"
      targetNamespace="http://sum.odws.univaq.it/" version="1.0">
      <xs:element name="sum" type="tns:sum"/>
      <xs:element name="sumResponse" type="tns:sumResponse"/>
      <xs:complexType name="sum">
        <xs:sequence>
          <xs:element name="arg0" type="xs:int"/>
          <xs:element name="arg1" type="xs:int"/>
        </xs:sequence>
      </xs:complexType>
      <xs:complexType name="sumResponse">
        <xs:sequence>
          <xs:element name="return" type="xs:int"/>
        </xs:sequence>
      </xs:complexType>
    </xs:schema>
  </wsdl:types>
  <wsdl:message name="sum">
    <wsdl:part element="tns:sum" name="parameters"></wsdl:part>
  </wsdl:message>
  <wsdl:message name="sumResponse">
    <wsdl:part element="tns:sumResponse" name="parameters"></wsdl:part>
  </wsdl:message>
  <wsdl:portType name="Sum">
    <wsdl:operation name="sum">
      <wsdl:input message="tns:sum" name="sum"></wsdl:input>
      <wsdl:output message="tns:sumResponse" name="sumResponse"></wsdl:output>
    </wsdl:operation>
  </wsdl:portType>
  <wsdl:binding name="SumImplServiceSoapBinding" type="tns:Sum">
    <soap:binding style="document" transport="http://schemas.xmlsoap.org/soap/http"/>
    <wsdl:operation name="sum">
      <soap:operation soapAction="" style="document"/>
      <wsdl:input name="sum">
        <soap:body use="literal"/>
      </wsdl:input>
      <wsdl:output name="sumResponse">
        <soap:body use="literal"/>
      </wsdl:output>
    </wsdl:operation>
  </wsdl:binding>
  <wsdl:service name="SumImplService">
    <wsdl:port binding="tns:SumImplServiceSoapBinding" name="SumImplPort">
      <soap:address location="http://localhost:8080/SumSOAPService/sum"/>
    </wsdl:port>
  </wsdl:service>
</wsdl:definitions>
```

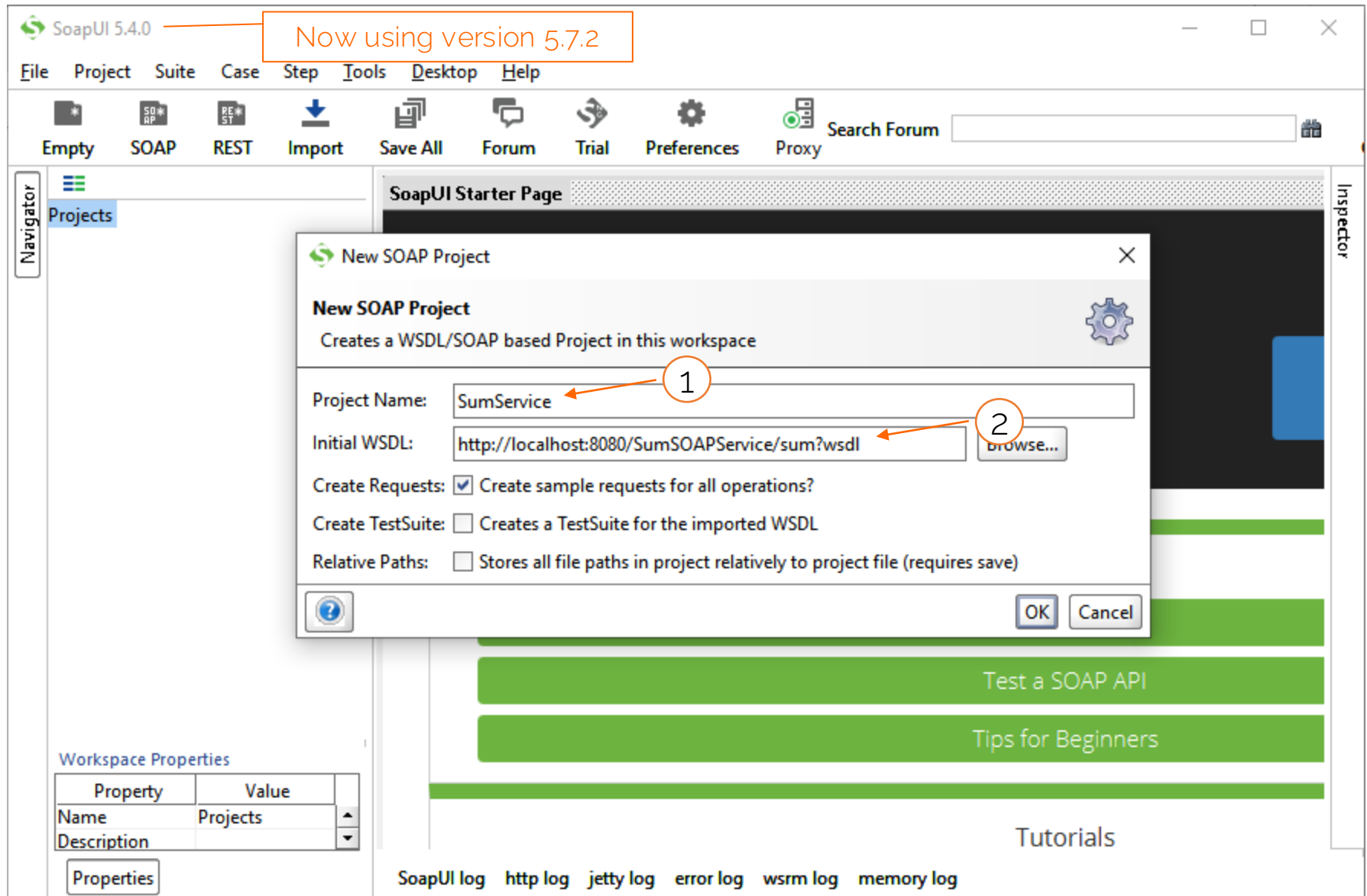
<http://localhost:8080/SumSOAPServiceNew/sum?wsdl>

How can we invoke our SOAP service?

Playing with SoapUI



Playing with SoapUI



Playing with SoapUI

SoapUI 5.4.0 — Now using version 5.7.2

File Project Suite Case Step Tools Desktop Help

Empty SOAP REST Import Save All Forum Trial Preferences Proxy Search Forum

Navigator

- projects
 - SumService
 - SumImplServiceSoapBinding
 - sum
 - Request 1

Request Properties

Property	Value
Name	Request 1
Description	

Properties

SoapUI Starter Page

Req 3

http://localhost:8080/SumSOAPService/sum

XML

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <sum:sum>
      <arg0>5</arg0>
      <arg1>5</arg1>
    </sum:sum>
  </soapenv:Body>
</soapenv:Envelope>
```

Raw

XML

```
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:sumResponse xmlns:ns2="http://schemas.xmlsoap.org/soap/envelope/">
      <return>10</return>
    </ns2:sumResponse>
  </soap:Body>
</soap:Envelope>
```

Header... Attachme... SSL I... WS... JMS...

response time: 519ms (202 bytes)

1 : 1

Tutorials

SoapUI log http log jetty log error log wsrm log memory log

Handling Exceptions Using SOAP Faults

Create the SumException class

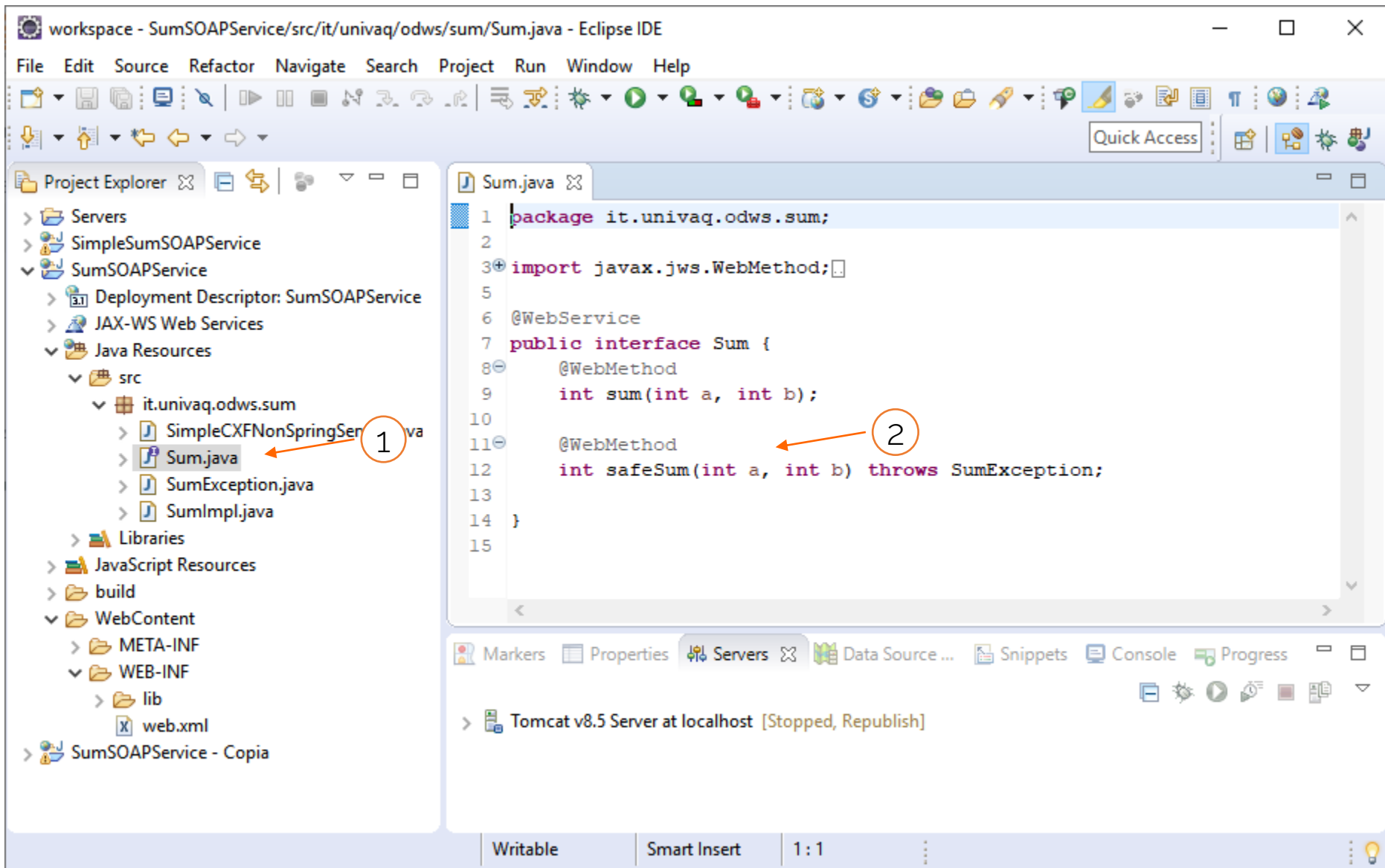
The screenshot shows the Eclipse IDE interface with the following components:

- Project Explorer (Left):** Displays the project structure. The path `src/it.univaq.odws.sum/SumException.java` is highlighted, indicated by an orange arrow and a circled '1'.
- Editor (Center):** Shows the code for `SumException.java`. The code is as follows:

```
1 package it.univaq.odws.sum;
2
3 public class SumException extends Exception {
4     private static final long serialVersionUID = 5113561641382114903L;
5
6     public SumException(String message) {
7         super(message);
8     }
9 }
10
11
```

Annotations in the code include:
 - An orange arrow and a circled '2' pointing to the line `public class SumException extends Exception {`.
 - An orange arrow and a circled '3' pointing to the line `super(message);` inside the constructor.
- Bottom Panel:** Contains tabs for Markers, Properties, Servers, Data Source, Snippets, Console, and Progress. The 'Servers' tab is active, showing 'Tomcat v8.5 Server at localhost [Stopped, Republish]'.

Add the safeSum operation



Implement the safeSum operation

The screenshot shows the Eclipse IDE interface. On the left, the Project Explorer displays the project structure, with `SumImpl.java` selected under the `src` directory. An orange arrow labeled '1' points to this file. The main editor window shows the code for `SumImpl.java`. The code defines a package `it.univaq.odws.sum` and a class `SumImpl` that implements the `Sum` interface. It includes two methods: `sum(int a, int b)` and `safeSum(int a, int b) throws SumException`. An orange arrow labeled '3' points to the `safeSum` method signature. The implementation of `safeSum` uses a try-catch block to call `Math.addExact(a, b)` and throw a `SumException` if an `ArithmeticException` occurs. An orange arrow labeled '4' points to the `throw new SumException(e.getMessage());` line. A text box on the right explains the purpose of the `safeSum` method.

```
1 package it.univaq.odws.sum;
2
3 public class SumImpl implements Sum {
4     @Override
5     public int sum(int a, int b) {
6         return a + b;
7     }
8
9     @Override
10    public int safeSum(int a, int b) throws SumException {
11        try {
12            return Math.addExact(a, b);
13        } catch (ArithmeticException e) {
14            throw new SumException(e.getMessage());
15        }
16    }
17 }
```

We now invoke the static method on Math that throws an ArithmeticException if the result overflows an integer

Deploy the SumSOAPService

CXF - Service list

localhost:8080/SumSOAPService/

Available SOAP services:

Sum	Endpoint address: http://localhost:8080/SumSOAPService/sum WSDL : http://sum.odws.univaq.it/SumImplService Target namespace: http://sum.odws.univaq.it/
- sum - safeSum	

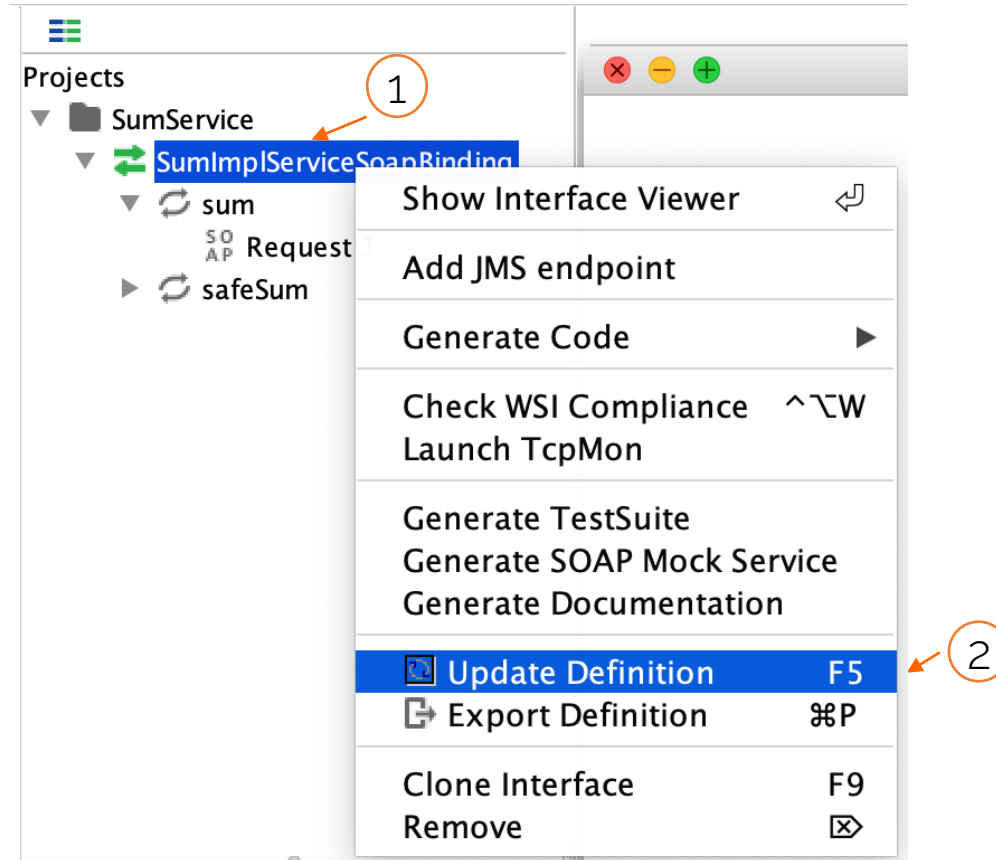
1

```
<?xml version='1.0'?>
<wsdl:portType name="Sum">
  <wsdl:operation name="sum">
    <wsdl:input message="tns:sum" name="sum"></wsdl:input>
    <wsdl:output message="tns:sumResponse" name="sumResponse"></wsdl:output>
  </wsdl:operation>
  <wsdl:operation name="safeSum">
    <wsdl:input message="tns:safeSum" name="safeSum"></wsdl:input>
    <wsdl:output message="tns:safeSumResponse" name="safeSumResponse"></wsdl:output>
    <wsdl:fault message="tns:SumException" name="SumException"></wsdl:fault>
  </wsdl:operation>
</wsdl:portType>
```

2

3

Refresh to get the newly added operation



Playing with the WSDL Fault Messages

The screenshot shows the SoapUI 5.4.0 interface. On the left, the **Navigator** pane shows a project named **SumService** with a sub-project **SumImplService**. Inside **SumImplService**, there are two SOAP API requests: **safeSum** (circled with a red '1') and **sum**. The **Request Properties** pane at the bottom left shows the selected request is **Request 1**, with message **308**, encoding **UTF-8**, and endpoint **http://lo...**.

The main window displays **Request 1** at the URL **http://localhost:8080/SumSOAPService/sum**. The **Raw XML** view shows the request body (circled with a red '2'):

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <sum:safeSum>
      <arg0>1073741824</arg0>
      <arg1>1073741824</arg1>
    </sum:safeSum>
  </soapenv:Body>
</soapenv:Envelope>
```

Below the XML, a text box contains the calculation: $1.073.741.824 + 1.073.741.824 =$ followed by a sad face emoji (😞).

The **Inspector** pane on the right shows the **Raw XML** view of the response, which is a fault (circled with a red '3'):

```
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <soap:Fault>
      <faultcode>soap:Server</faultcode>
      <faultstring>integer overflow</faultstring>
      <detail>
        <ns1:SumException xmlns:ns1="http://sum.c" />
      </detail>
    </soap:Fault>
  </soap:Body>
</soap:Envelope>
```

The status bar at the bottom shows **response time: 208ms (286 bytes)** and a list of logs: **SoapUI log**, **http log**, **jetty log**, **error log**, **wsm log**, and **memory log**.

Playing with the WSDL Fault Messages

The screenshot shows the SoapUI 5.4.0 interface. On the left, the 'Navigator' pane shows a project named 'SumService' with a sub-project 'SumImplService' containing two SOAP requests: 'safeSum' and 'sum'. The 'Request Properties' pane at the bottom left shows details for 'Request 1': Name, Message 308, Encoding UTF-8, and Endpoint http://localhost:8080/SumSOAPService/sum. The main window displays 'Request 1' with a URL bar showing the endpoint. The 'Raw XML' view shows the request body with two arguments, each with the value 1073741824. A red box highlights the calculation: 1.073.741.824 + 1.073.741.824 = 2.147.483.648 with a sad face emoji. The 'Inspector' pane on the right shows the fault response. A red box highlights the fault code 'soap:Server' and the fault string 'integer overflow'. Another red box highlights the value '2.147.483.647' in the fault details. A text box explains that this is the MAX_VALUE of an int, equal to 2³¹-1. A 'See Also' section points to 'Constant Field Values' and mentions the MIN_VALUE is -2³¹ = -2147483648. The bottom status bar shows 'response time: 208ms (286 bytes)' and a log area with various log types.

SoapUI 5.4.0

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Navigator

Projects

- SumService
 - SumImplService
 - safeSum
 - sum

Request Properties

Property	Value
Name	Request 1
Descript...	
Message...	308
Encoding	UTF-8
Endpoint	http://lo...

Request 1

http://localhost:8080/SumSOAPService/sum

Raw XML

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <sum:safeSum>
      <arg0>1073741824</arg0>
      <arg1>1073741824</arg1>
    </sum:safeSum>
  </soapenv:Body>
</soapenv:Envelope>
```

1.073.741.824 +
1.073.741.824 =
2.147.483.648 ☹️

Raw XML

```
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <soap:Fault>
      <faultcode>soap:Server</faultcode>
      <faultstring>integer overflow</faultstring>
      <detail>
        <ns1:SumException xmlns:ns1="http://sum.c...>
        </ns1:SumException>
      </detail>
    </soap:Fault>
  </soap:Body>
</soap:Envelope>
```

Inspector

MAX_VALUE

2.147.483.647

@Native

public static final int MAX_VALUE

A constant holding the maximum value an int can have, 2³¹-1.

See Also:

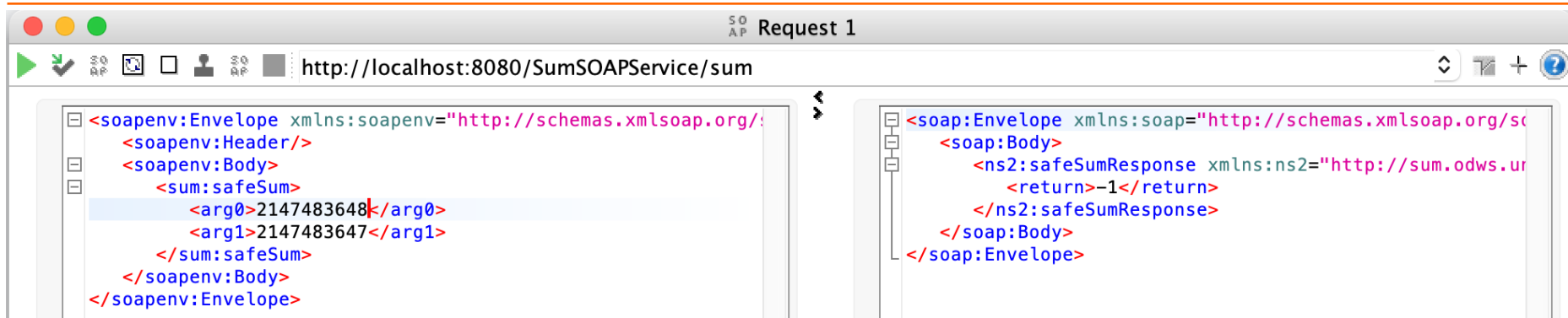
Constant Field Values

The MIN_VALUE is
-2³¹ = -2147483648

response time: 208ms (286 bytes)

SoapUI log http log jetty log error log wsrm log memory log

A curios case



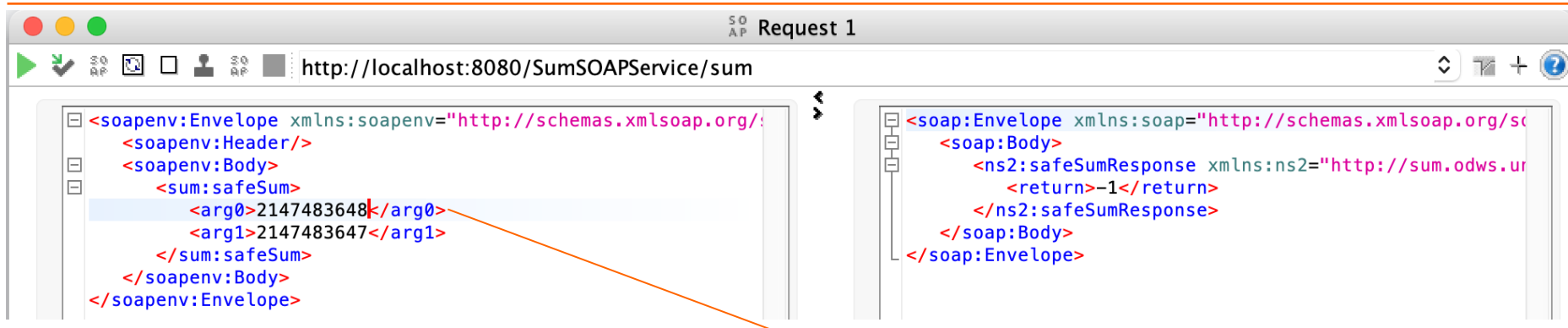
The screenshot shows a SOAP client interface with a title bar "SOAP Request 1". The address bar displays "http://localhost:8080/SumSOAPService/sum". The interface is split into two panels. The left panel shows the SOAP request XML, and the right panel shows the SOAP response XML.

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <sum:safeSum>
      <arg0>2147483648</arg0>
      <arg1>2147483647</arg1>
    </sum:safeSum>
  </soapenv:Body>
</soapenv:Envelope>
```

```
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:safeSumResponse xmlns:ns2="http://sum.odws.univ-lorraine.fr/sum" >
      <return>-1</return>
    </ns2:safeSumResponse>
  </soap:Body>
</soap:Envelope>
```



An interesting case



SOAP Request 1

http://localhost:8080/SumSOAPService/sum

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <sum:safeSum>
      <arg0>2147483648</arg0>
      <arg1>2147483647</arg1>
    </sum:safeSum>
  </soapenv:Body>
</soapenv:Envelope>
```

```
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:safeSumResponse xmlns:ns2="http://sum.odws.org/sum">
      <return>-1</return>
    </ns2:safeSumResponse>
  </soap:Body>
</soap:Envelope>
```

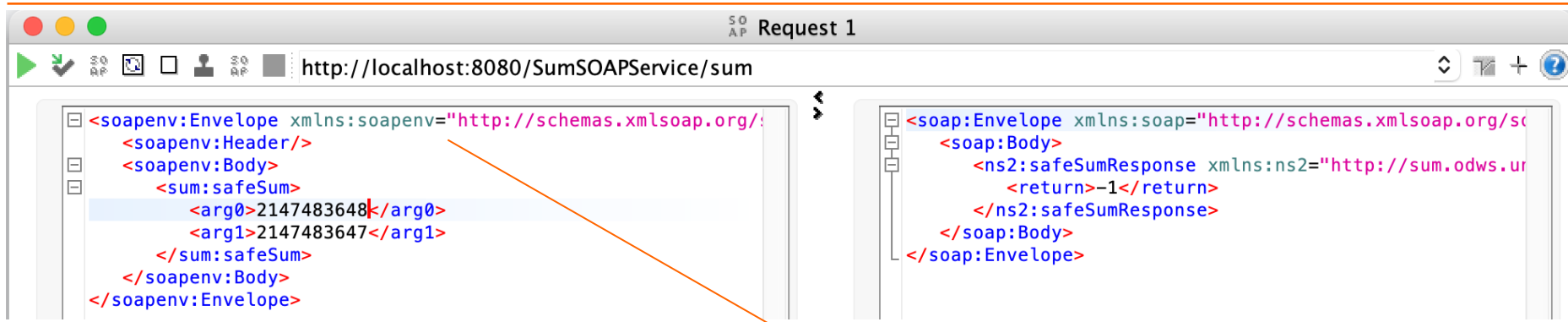
That's a different problem

- Here, the single argument `arg0` **overflows** (by one in our example), thus, due to marshalling and unmarshalling problems, it **goes back to the minimum value** ($2^{31} = -2.147.483.648$) and continues from there:

$$-2.147.483.648 + 2.147.483.647 = -1 \text{ ☹️}$$

- Should it **underflows**? → next slide 

A interesting case



That's a different problem

- Here, the single argument `arg0` **overflows** (by one in our example), thus, due to marshalling and unmarshalling problems, it **goes back to the minimum value** ($2^{31} = -2.147.483.648$) and continues from there:

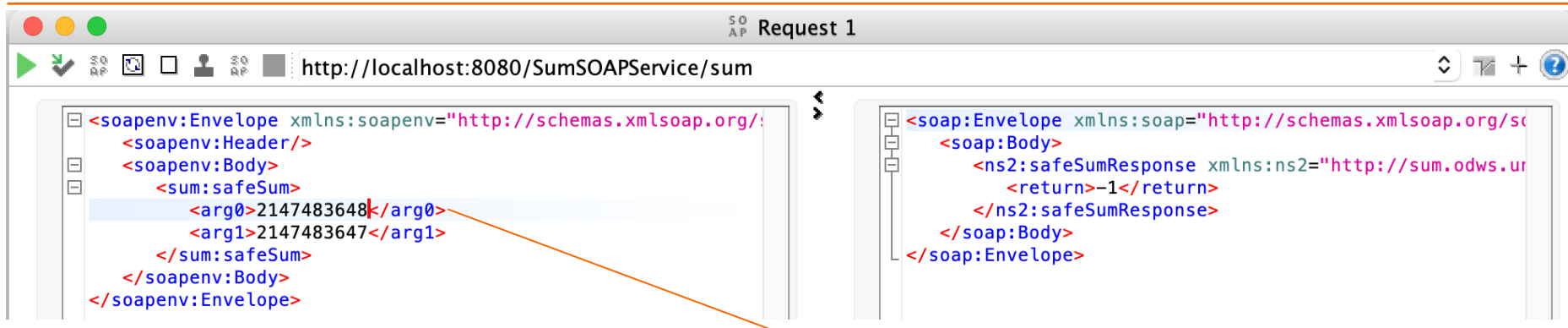
$$-2.147.483.648 + 2.147.483.647 = -1 \text{ 😞}$$

- Should it **underflows** to **-2.147.483.649**, it goes back to the maximum value (**2.147.483.647**) and continues from there)

- What's the point here?



An interesting case



That's a different problem

- Here, the single argument `arg0` **overflows** (by one in our example), thus, due to marshalling and unmarshalling problems, it **goes back to the minimum value** ($2^{31} = -2.147.483.648$) and continues from there:

$$-2.147.483.648 + 2.147.483.647 = -1 \text{ ☹}$$

- Should it **underflows** to $-2.147.483.649$ it goes back to the maximum value ($2.147.483.647$) and continues from there)

Note that, at the Java programming level, you would get an "immediate error" as you type numbers that overflow or underflow... Thus, the problem can be solved beforehand



The literal 2147483648 of type int is out of range

```
int a = Math.addExact(2147483648, 2);
```



The literal 2147483649 of type int is out of range

```
int a = Math.addExact(-2147483649, 2);
```

Playing with the WSDL Fault Messages

The screenshot shows the SoapUI 5.4.0 interface. On the left, the 'Navigator' pane shows a project named 'SumService' with a sub-project 'SumImplService' containing two SOAP API endpoints: 'safeSum' and 'sum'. The 'Request Properties' pane at the bottom left shows the selected request 'Request 1' with a message ID of 308, UTF-8 encoding, and endpoint 'http://localhost:8080/SumSOAPService/sum'. The main window displays the raw XML of the request and response. The request is a SOAP envelope with a body containing a 'sum:safeSum' operation with two arguments, both set to '1073741824'. The response is a SOAP fault with a fault code of 'soap:server' and a fault string of 'integer overflow'. A large orange question mark is drawn over the fault string. A red speech bubble with a sad face icon contains the text 'Try also to invoke the (unsafe) sum operation ...'. The status bar at the bottom shows 'response time: 208ms (286 bytes)' and various log files.

SoapUI 5.4.0

File Project Suite Case Step Tools Desktop Help

Empty SOAP REST Import Save All Forum Trial Preferences Proxy Search Forum Online

Navigator

Projects

- SumService
 - SumImplService
 - safeSum
 - SOAP API Request
 - sum

Request Properties

Property	Value
Name	Request 1
Descript...	
Message...	308
Encoding	UTF-8
Endpoint	http://lo...

Request 1

http://localhost:8080/SumSOAPService/sum

Raw XML

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <sum:safeSum>
      <arg0>1073741824</arg0>
      <arg1>1073741824</arg1>
    </sum:safeSum>
  </soapenv:Body>
</soapenv:Envelope>
```

Raw XML

```
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <soap:Fault>
      <faultcode>soap:server</faultcode>
      <faultstring>integer overflow</faultstring>
      <detail>
        <ns1:SumException xmlns:ns1="http://sum.c...>
        </ns1:SumException>
      </detail>
    </soap:Fault>
  </soap:Body>
</soap:Envelope>
```

Try also to invoke the (unsafe) sum operation ...

response time: 208ms (286 bytes)

SoapUI log http log jetty log error log wsrm log memory log

Playing with the WSDL Fault Messages

SOAP Fault can be a **modeled fault** or an **un-modeled fault**

- **Modeled fault** is a **user defined fault** message when an **exception is caught** in the code and sent as a part of SOAP Fault, wrapped inside the element `<detail/>`

The screenshot shows a SOAP client window titled "Request 1" with the URL `http://localhost:8080/SumSOAPService/sum`. The left pane displays the request XML, and the right pane displays the response XML. The response is a SOAP fault with a detailed exception.

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <sum:safeSum>
      <arg0>2147483647</arg0>
      <arg1>2147483647</arg1>
    </sum:safeSum>
  </soapenv:Body>
</soapenv:Envelope>
```

```
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <soap:Fault>
      <faultcode>soap:Server</faultcode>
      <faultstring>integer overflow</faultstring>
      <detail>
        <ns1:SumException xmlns:ns1="http://sum.odws.univaq.it/">
        </ns1:SumException>
      </detail>
    </soap:Fault>
  </soap:Body>
</soap:Envelope>
```

- **Un-modeled fault** is a **system generated fault** which is sent when some **unexpected/unhandled exception occurs**, which looks like the one below

The screenshot shows a SOAP client window titled "Request 1" with the URL `http://localhost:8080/SumSOAPService/sum`. The left pane displays the request XML, and the right pane displays the response XML. The response is a SOAP fault without a detailed exception.

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <sum:safeSum>
      <arg0>2147483647</arg0>
      <arg1>2147483647</arg1>
    </sum:safeSum>
  </soapenv:Body>
</soapenv:Envelope>
```

```
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <soap:Fault>
      <faultcode>soap:Server</faultcode>
      <faultstring>integer overflow</faultstring>
    </soap:Fault>
  </soap:Body>
</soap:Envelope>
```

How to get an un-modeled fault

The screenshot shows an IDE with a Project Explorer on the left and a code editor on the right. The Project Explorer shows a project structure with a package `it.univaq.odws.sum` containing files `SimpleCXFNonSpringServlet.java`, `Sum.java`, `SumException.java`, and `SumImpl.java`. The code editor shows the `SumImpl.java` file. It contains two methods: `sum(int a, int b)` and `safeSum(int a, int b) throws SumException`. The `safeSum` method has a try-catch block that catches `ArithmeticException` and throws `SumException`. A red circle with the number '1' is drawn around the closing brace of the try block in the `safeSum` method, indicating that the exception is not caught. An orange arrow points from the text 'without catching the exception...' to this circle. The text 'without catching the exception...' is written in orange inside a tilted orange box at the bottom right of the code editor.

```
1 package it.univaq.odws.sum;
2
3 public class SumImpl implements Sum {
4
5     @Override
6     public int sum(int a, int b) {
7         return a + b;
8     }
9
10    @Override
11    public int safeSum(int a, int b) throws SumException {
12
13        // try {
14            return Math.addExact(a, b);
15        // } catch (ArithmeticException e) {
16        //     throw new SumException(e.getMessage());
17        // }
18
19    }
20
21 }
22
```

without catching the exception...

More on faults later...

The Contract First Approach

WS development approaches

Bottom-up or Code First (Java → WSDL)

- It is about creating web services by starting from the Java classes, annotating them, and generating the corresponding WSDL

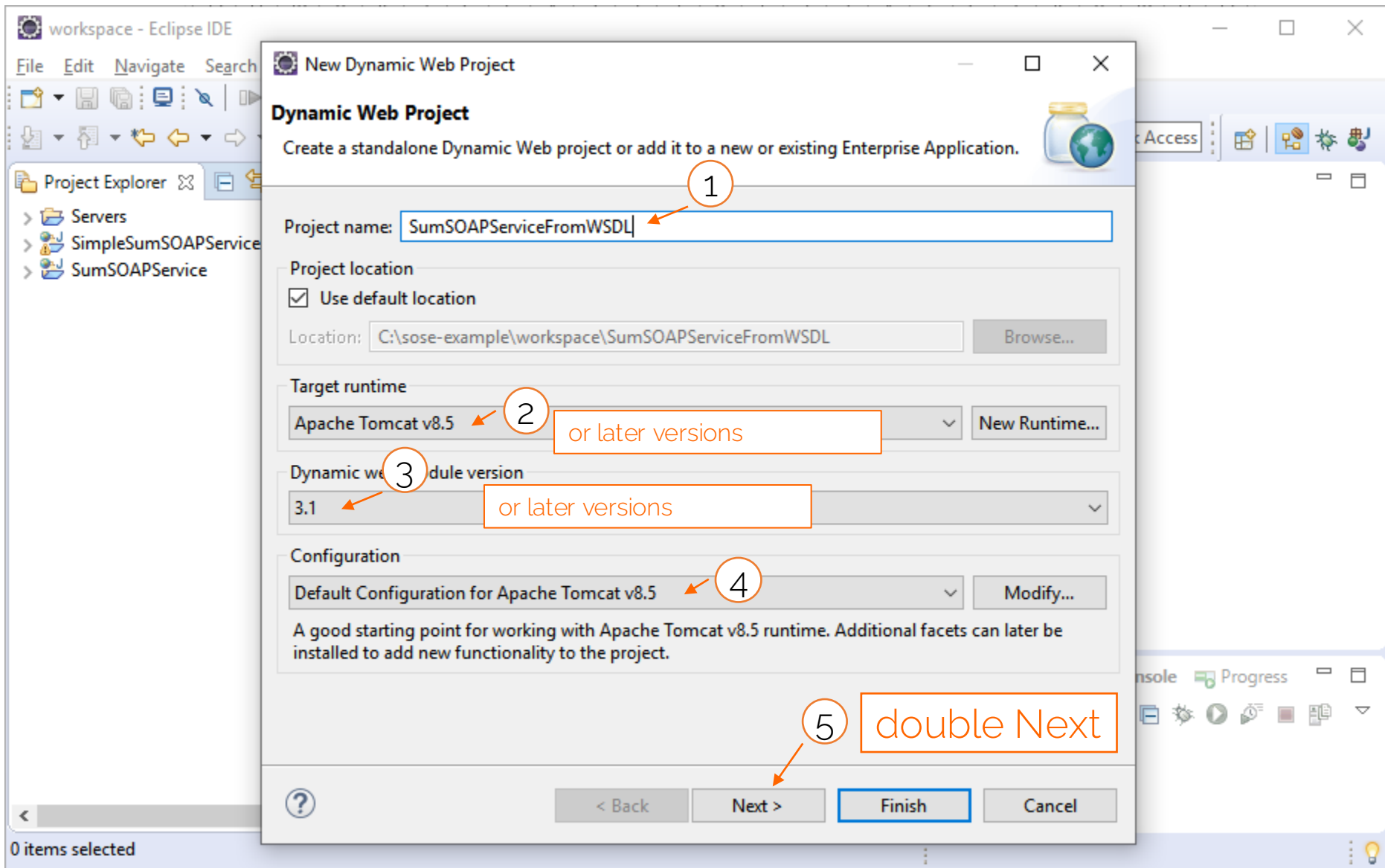
Top-Down or Contract First Approach (WSDL → Java)

- It is about creating web services by starting from the WSDL and generating the corresponding Java classes (skeleton classes)

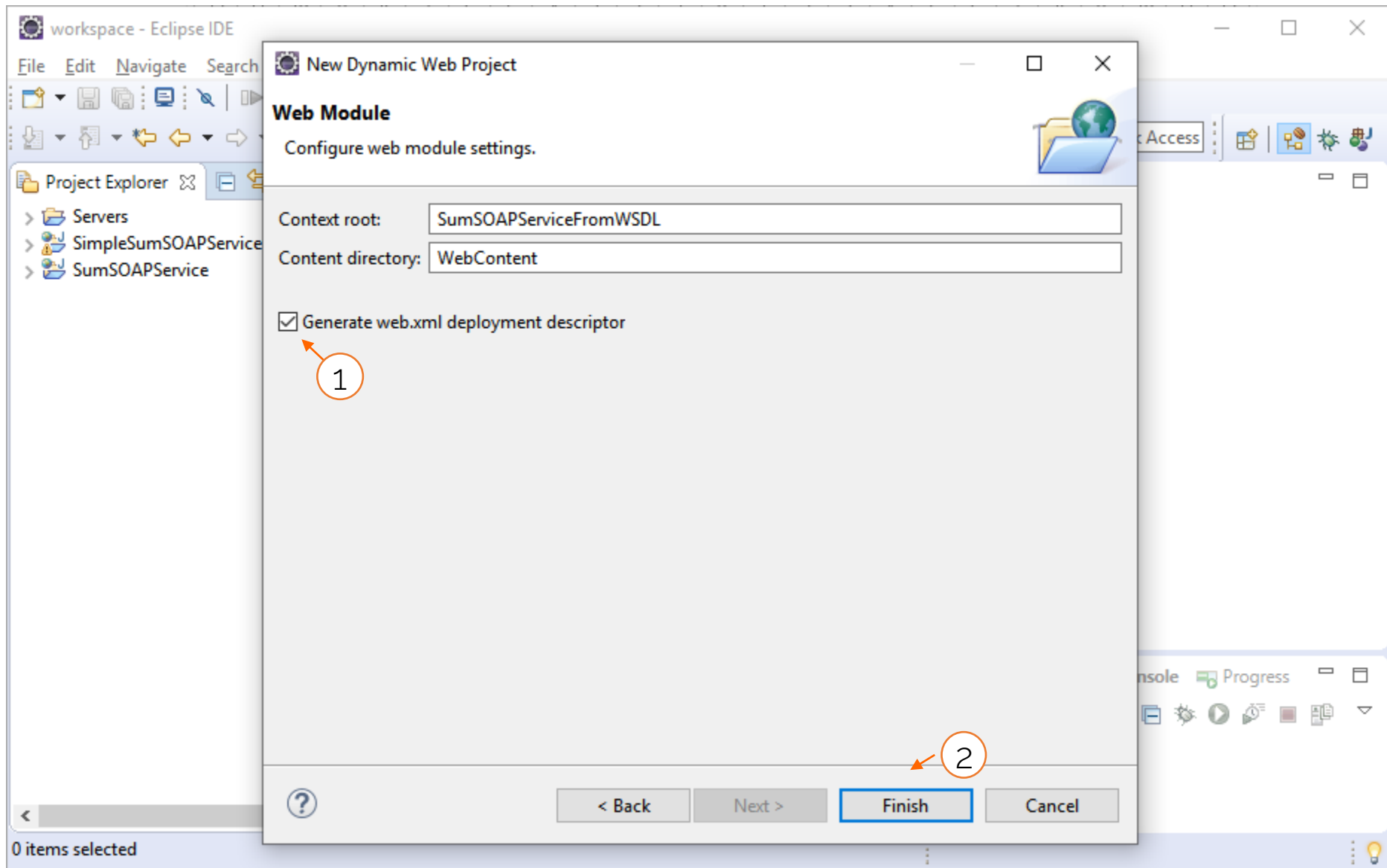
Hybrid

- Start with code first approach by creating Java classes and thereby generating WSDL files. Customize the generated WSDL files, then WSDL → Java, or viceversa

Create the SumSOAPServiceFromWSDL



Create the SumSOAPServiceFromWSDL



Define the WSDL

workspace - SumSOAPServiceFromWSDL/WebContent/WEB-INF/sum.wsdl - Eclipse IDE

File Edit Source Refactor Navigate Search Project WSDL Editor Run Window Help

Project Explorer

- Servers
- SimpleSumSOAPService
- SumSOAPService
- SumSOAPServiceFromWSDL
 - Deployment Descriptor: SumSOAPServiceFr
 - JAX-WS Web Services
 - Java Resources
 - src
 - Libraries
 - JavaScript Resources
 - build
 - WebContent
 - META-INF
 - WEB-INF
 - lib
 - sum.wsdl
 - web.xml
 - JSR-109 Web Services

sum.wsdl

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <wsdl:definitions
3     xmlns:xsd="http://www.w3.org/2001/XMLSchema"
4     xmlns:wsdl="http://schemas.xmlsoap.org/wsdl/"
5     xmlns:tns="http://sum.odws.univaq.it/"
6     xmlns:soap="http://schemas.xmlsoap.org/wsdl/soap/"
7     xmlns:ns1="http://schemas.xmlsoap.org/soap/http" name="SumImplSei
8     targetNamespace="http://sum.odws.univaq.it/">
9     <wsdl:types>
10     <wsdl:message name="safeSum">
11     <wsdl:message name="SumException">
12     <wsdl:message name="sum">
13     <wsdl:message name="safeSumResponse">
14     <wsdl:message name="sumResponse">
15     <wsdl:portType name="Sum">
```

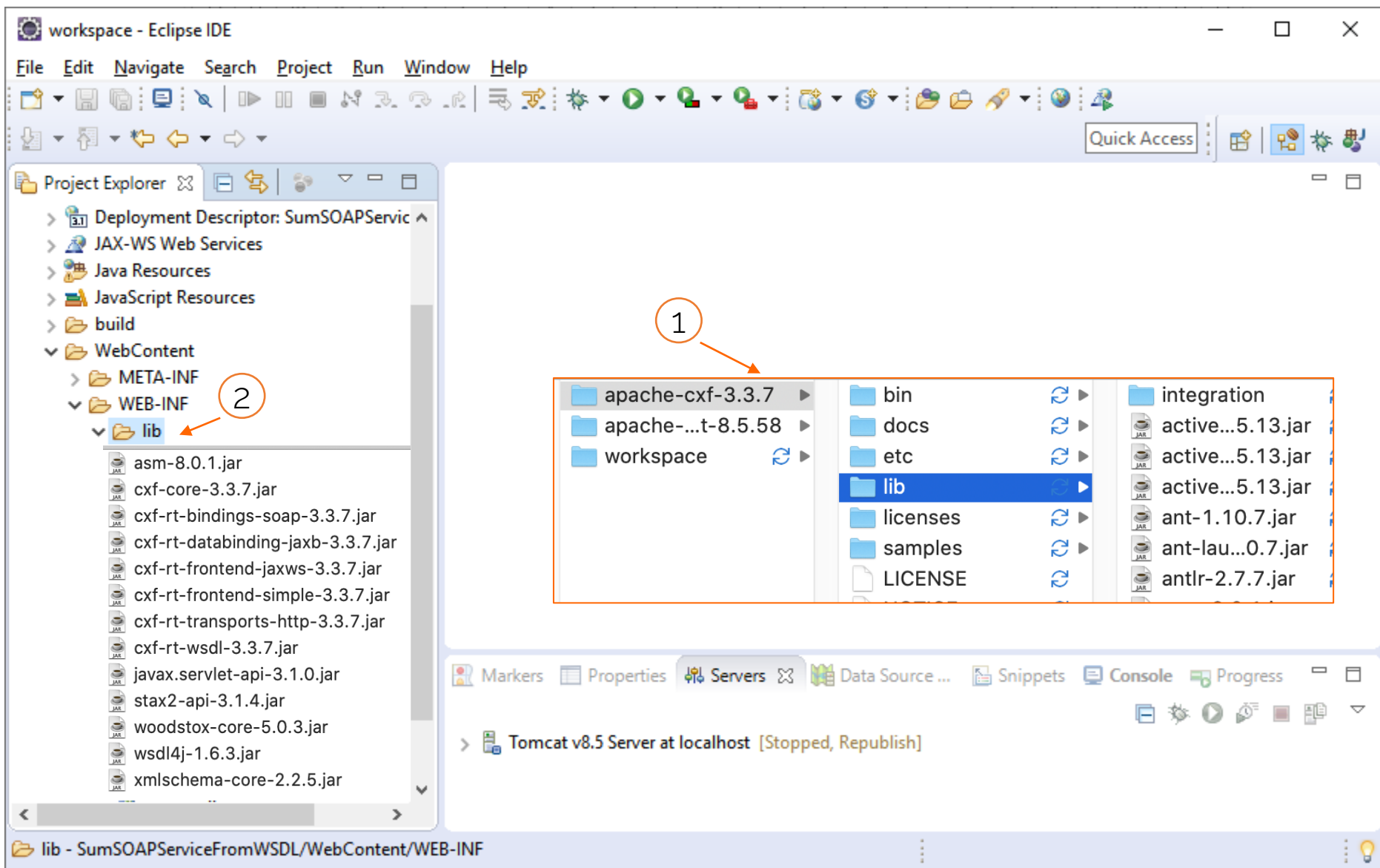
1 It is the SumSOAPService WSDL

Properties Servers Data Source ... Snippets Console Progress

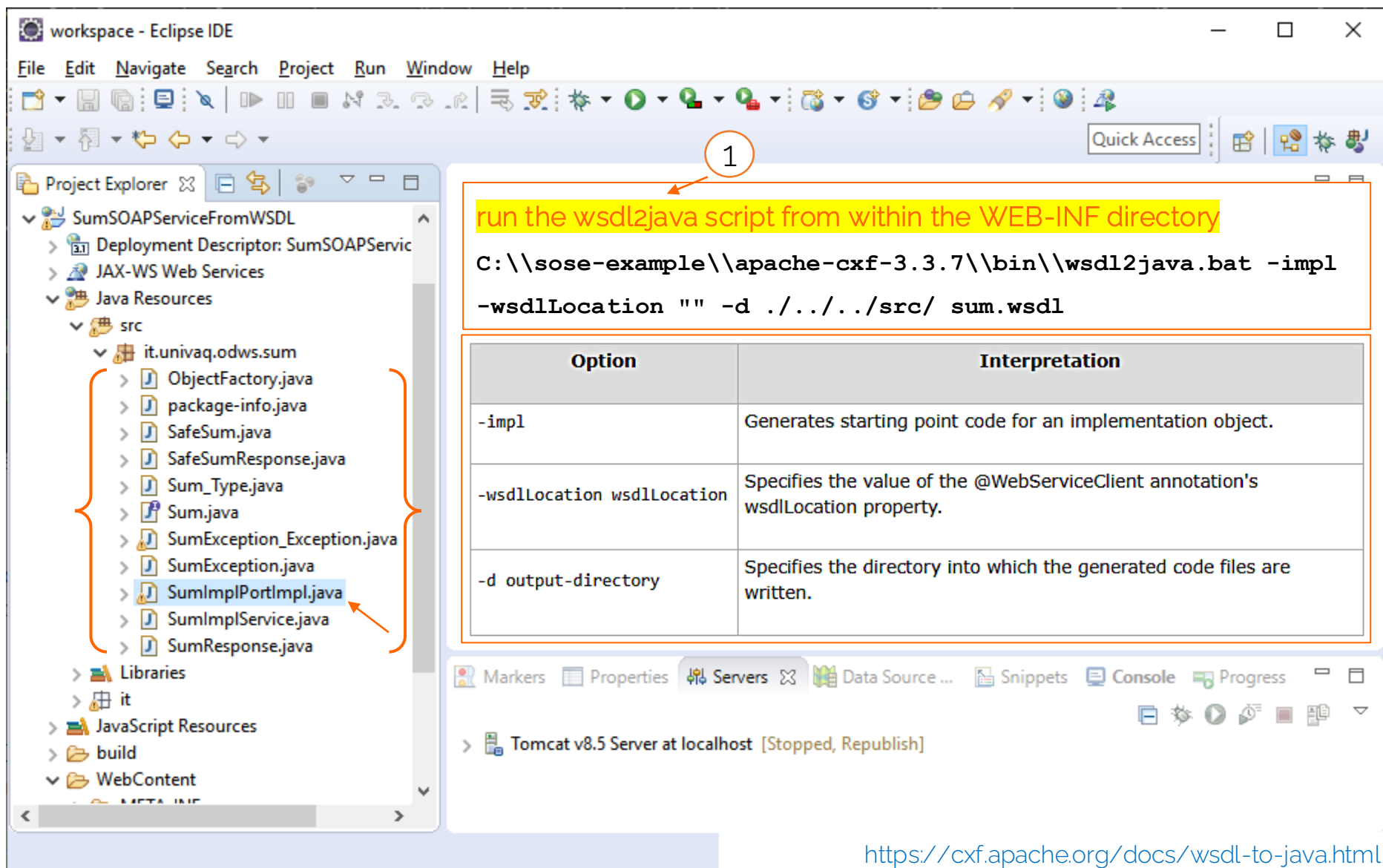
Tomcat v8.5 Server at localhost [Stopped, Republish]

Writable Smart Insert 1:1

Put CXF jars in lib folder (remember, not all jars...)



Execute the wsdl2java script (on Window)



workspace - Eclipse IDE

File Edit Navigate Search Project Run Window Help

Project Explorer

- SumSOAPServiceFromWSDL
 - Deployment Descriptor: SumSOAPService
 - JAX-WS Web Services
 - Java Resources
 - src
 - it.univaq.odws.sum
 - ObjectFactory.java
 - package-info.java
 - SafeSum.java
 - SafeSumResponse.java
 - Sum_Type.java
 - Sum.java
 - SumException_Exception.java
 - SumException.java
 - SumImplPortImpl.java
 - SumImplService.java
 - SumResponse.java
 - Libraries
 - it
 - JavaScript Resources
 - build
 - WebContent

1

run the wsdl2java script from within the WEB-INF directory

```
C:\sese-example\apache-cxf-3.3.7\bin\wsdl2java.bat -impl -wsdlLocation "" -d ../../../../src/ sum.wsdl
```

Option	Interpretation
-impl	Generates starting point code for an implementation object.
-wsdlLocation wsdlLocation	Specifies the value of the @WebServiceClient annotation's wsdlLocation property.
-d output-directory	Specifies the directory into which the generated code files are written.

Markers Properties Servers Data Source ... Snippets Console Progress

Tomcat v8.5 Server at localhost [Stopped, Republish]

<https://cxf.apache.org/docs/wsdl-to-java.html>

Execute the wsdl2java script (on my MacOS)

```
src — -zsh — 108x35

marcoautili@Marcos-MBP-2020 src %
marcoautili@Marcos-MBP-2020 src % cd /Applications/java/eclipse-workspace/odws/SumSOAPServiceFromWSDL/src
marcoautili@Marcos-MBP-2020 src %
marcoautili@Marcos-MBP-2020 src % ls
marcoautili@Marcos-MBP-2020 src %
marcoautili@Marcos-MBP-2020 src % cd /Applications/java/eclipse-workspace/odws/SumSOAPServiceFromWSDL/WebContent/WEB-INF
marcoautili@Marcos-MBP-2020 WEB-INF %
marcoautili@Marcos-MBP-2020 WEB-INF %
marcoautili@Marcos-MBP-2020 WEB-INF % ls
lib          sum.wsdl      web.xml
marcoautili@Marcos-MBP-2020 WEB-INF %
marcoautili@Marcos-MBP-2020 WEB-INF %
marcoautili@Marcos-MBP-2020 WEB-INF % export JAVA_HOME=/Library/Java/JavaVirtualMachines/jdk1.8.0_291.jdk/Contents/Home
marcoautili@Marcos-MBP-2020 WEB-INF %
marcoautili@Marcos-MBP-2020 WEB-INF %
marcoautili@Marcos-MBP-2020 WEB-INF % echo $JAVA_HOME
/Library/Java/JavaVirtualMachines/jdk1.8.0_291.jdk/Contents/Home
marcoautili@Marcos-MBP-2020 WEB-INF %
marcoautili@Marcos-MBP-2020 WEB-INF %
marcoautili@Marcos-MBP-2020 WEB-INF % . /Applications/java/apache-cxf-3.4.3/bin/wsdl2java -impl -wsdlLocation "" -d ../../src/ sum.wsdl
marcoautili@Marcos-MBP-2020 WEB-INF %
marcoautili@Marcos-MBP-2020 WEB-INF %
marcoautili@Marcos-MBP-2020 WEB-INF % cd /Applications/java/eclipse-workspace/odws/SumSOAPServiceFromWSDL/src
marcoautili@Marcos-MBP-2020 src %
marcoautili@Marcos-MBP-2020 src %
marcoautili@Marcos-MBP-2020 src % ls
it
marcoautili@Marcos-MBP-2020 src %
marcoautili@Marcos-MBP-2020 src %
```

The **src** folder is initially empty

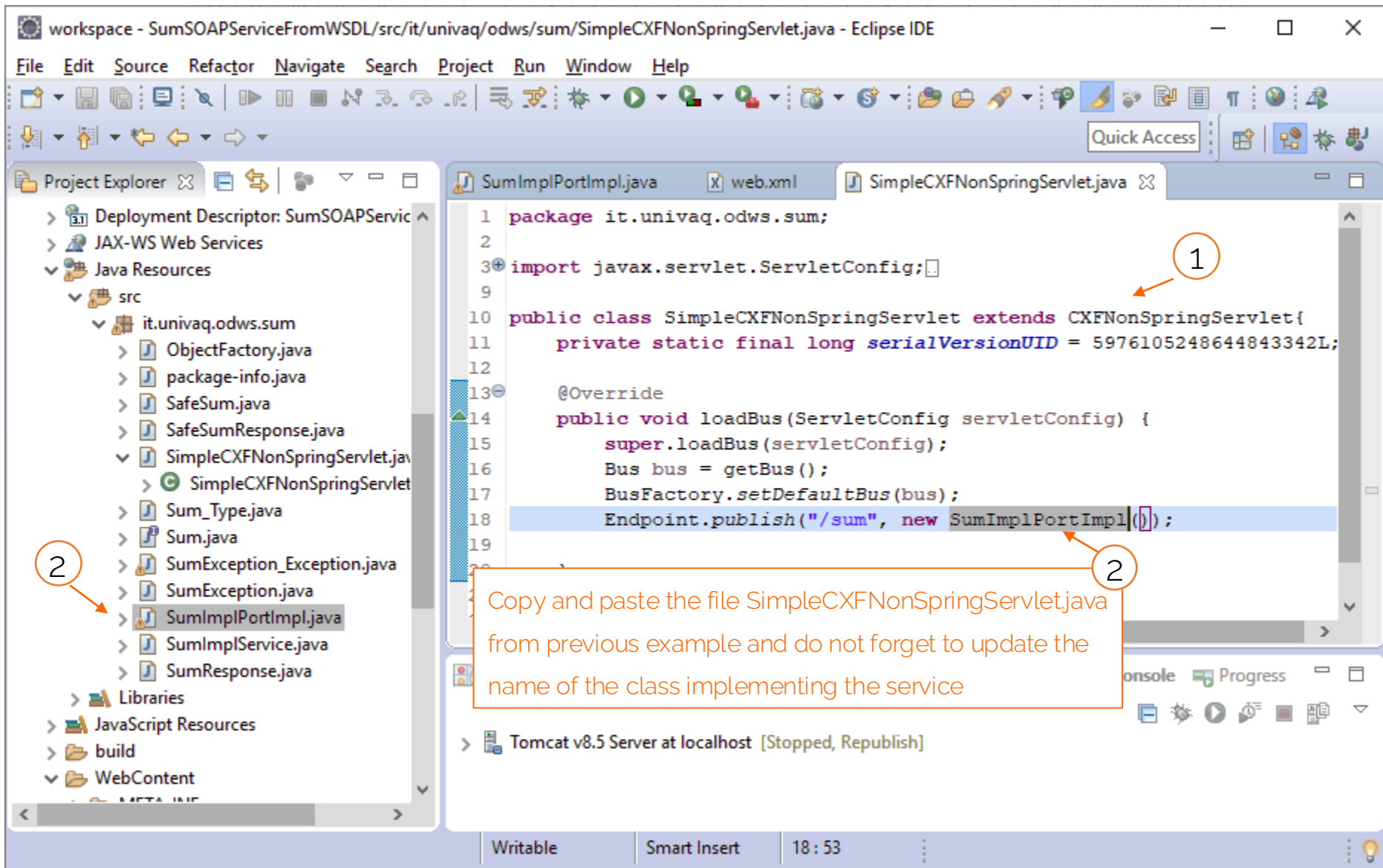
The **WEB-INF** folder contains the **sum.wsdl** file

Set the **JAVA_HOME** variable for terminal session only
(needed only if you have not set it permanently yet)

Do not forget to run this script by prefixing "**source**" or "**."**"

The **src** folder now contains the generated package... refresh and navigate it on Eclipse

Create the CXFNonSpringServlet



The screenshot shows the Eclipse IDE workspace for a project named 'SumSOAPServiceFromWSDL'. The Project Explorer on the left shows the package structure 'it.univaq.odws.sum'. The main editor displays the code for 'SimpleCXFNonSpringServlet.java'. The code includes package declarations, imports, and a class that extends 'CXFNonSpringServlet'. A text box with an arrow points to the class name in the code, and another arrow points to the file in the Project Explorer.

workspace - SumSOAPServiceFromWSDL/src/it/univaq/odws/sum/SimpleCXFNonSpringServlet.java - Eclipse IDE

File Edit Source Refactor Navigate Search Project Run Window Help

Project Explorer

- Deployment Descriptor: SumSOAPService
- JAX-WS Web Services
- Java Resources
 - src
 - it.univaq.odws.sum
 - ObjectFactory.java
 - package-info.java
 - SafeSum.java
 - SafeSumResponse.java
 - SimpleCXFNonSpringServlet.java
 - SimpleCXFNonSpringServlet
 - Sum_Type.java
 - Sum.java
 - SumException_Exception.java
 - SumException.java
 - SumImplPortImpl.java
 - SumImplService.java
 - SumResponse.java
 - Libraries
 - JavaScript Resources
 - build
 - WebContent

```
1 package it.univaq.odws.sum;
2
3 import javax.servlet.ServletConfig;
4
5
6
7
8
9
10 public class SimpleCXFNonSpringServlet extends CXFNonSpringServlet{
11     private static final long serialVersionUID = 5976105248644843342L;
12
13     @Override
14     public void loadBus(ServletConfig servletConfig) {
15         super.loadBus(servletConfig);
16         Bus bus = getBus();
17         BusFactory.setDefaultBus(bus);
18         Endpoint.publish("/sum", new SumImplPortImpl());
19     }
20 }
```

Copy and paste the file SimpleCXFNonSpringServlet.java from previous example and do not forget to update the name of the class implementing the service

Tomcat v8.5 Server at localhost [Stopped, Republish]

Configure the servlet in the web.xml file

workspace - SumSOAPServiceFromWSDL/WebContent/WEB-INF/web.xml - Eclipse IDE

File Edit Navigate Search Project Run Design Window Help

Project Explorer

- Servers
- SimpleSumSOAPService
- SumSOAPService
- SumSOAPServiceFromWSDL
 - Deployment Descriptor: SumSOAPServiceFr
 - JAX-WS Web Services
 - Java Resources
 - src
 - it.univaq.odws.sum
 - Libraries
 - JavaScript Resources
 - build
 - WebContent
 - META-INF
 - WEB-INF
 - lib
 - sum.wsdl
 - web.xml
 - JSR-109 Web Services

SumImplPortImpl.java web.xml

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <web-app xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns='
3   <display-name>SumSOAPServiceFromWSDL</display-name>
4
5   <servlet>
6     <servlet-name>SimpleCXFNonSpringServlet</servlet-name>
7     <servlet-class>it.univaq.odws.sum.SimpleCXFNonSpringServlet</servl
8   </servlet>
9
10  <servlet-mapping>
11    <servlet-name>SimpleCXFNonSpringServlet</servlet-name>
12    <url-pattern>/*</url-pattern>
13  </servlet-mapping>
14
15  <welcome-file-list>
```

Design Source

Markers Properties Servers Data Source ... Snippets Console Progress

Tomcat v8.5 Server at localhost [Stopped, Republish]

0 items selected

Implement the sum and safeSum operations

The screenshot shows the Eclipse IDE interface with the following components:

- Project Explorer (Left):** Displays the project structure. The package `it.univaq.odws.sum` is expanded, and `SumImplPortImpl.java` is selected. A red circle with the number 1 points to this file.
- Editor (Center):** Shows the source code of `SumImplPortImpl.java`. The code includes:
 - A `sum` method (lines 38-41) that logs and returns the sum of two integers. A red circle with the number 2 points to the `return` statement.
 - A `safeSum` method (lines 46-58) that uses `Math.addExact` and catches `ArithmeticException` to throw a `SumException_Exception`. A red circle with the number 3 points to the `new SumException()` instantiation.
- Bottom Panel:** Includes tabs for Markers, Properties, Servers, Data Source Explorer, Snippets, and Console. The Servers tab is active, showing "Tomcat v8.5 Server at localhost [Stopped, Republish]".

Run the Service on Server

The screenshot shows the Eclipse IDE interface with the following components:

- Project Explorer:** Displays the project structure for `workspace - http://localhost:8080/SumSOAPServiceFromWSDL/`. The `SumSOAPServiceFromWSDL` project is expanded, showing sub-projects like `Deployment Descriptor: SumSOAPServiceFr`, `JAX-WS Web Services`, `Java Resources` (containing `src` with `it.univaq.odws.sum`), `Libraries`, `JavaScript Resources`, `build`, `WebContent` (containing `META-INF`, `WEB-INF` with `lib`), `sum.wsdl`, and `web.xml`. It also shows `JSR-109 Web Services`.
- Service List:** A tab titled "CXF - Service list" is active, showing the URL `http://localhost:8080/SumSOAPServiceFromWSDL/`. Below the URL, it lists "Available SOAP services:" in a table:

Sum	Endpoint address:
• sum	<code>http://localhost:8080/SumSOAPServiceFromWSDL/sum</code>
• safeSum	<code>WSDL : {http://sum.odws.univaq.it}SumImplService</code>
	<code>Target namespace: http://sum.odws.univaq.it/</code>

An orange circle with the number "1" and an arrow points to the `WSDL` entry in the table.
- Console:** Shows the output of the Tomcat server: `Tomcat v8.5 Server at localhost [Apache Tomcat] C:\Program Files\Java\jre1.8.0_221\bin\javaw.exe (3 ott 2019, 18:39:25)`.
- Status Bar:** Displays the word "Fine" on the left and a lightbulb icon on the right.

Find the differences

Computed Diff - Diff Checker

diffchecker.com/diff

Diffchecker

Home Images PDF Contact CLI Desktop

Sign in Create an account



Make it with Adobe Stock.
Get 10 free images >

Limited time offer: Get 10 free Adobe Stock images.

ads via Carbon

Recent Diffs Clear

Unsaved diff

Recent diffs are deleted on refresh

Saved Diffs

You must [log in](#) to save diffs

SumSOAPService

```
33 <xs:element minOccurs="0" name="sum" type="xs:string"/>
34 </xs:sequence>
35 </xs:complexType>
36 </xs:schema>
37 </wsdl:types>
38 <wsdl:message name="safeSum">
39   <wsdl:part element="tns:safeSum" name="parameters">
40   </wsdl:part>
41 </wsdl:message>
42 <wsdl:message name="SumException">
43   <wsdl:part element="tns:SumException" name="SumException">
44   </wsdl:part>
45 </wsdl:message>
46 <wsdl:message name="sum">
47   <wsdl:part element="tns:sum" name="parameters">
```

Editor

SumSOAPServiceFromWSDL

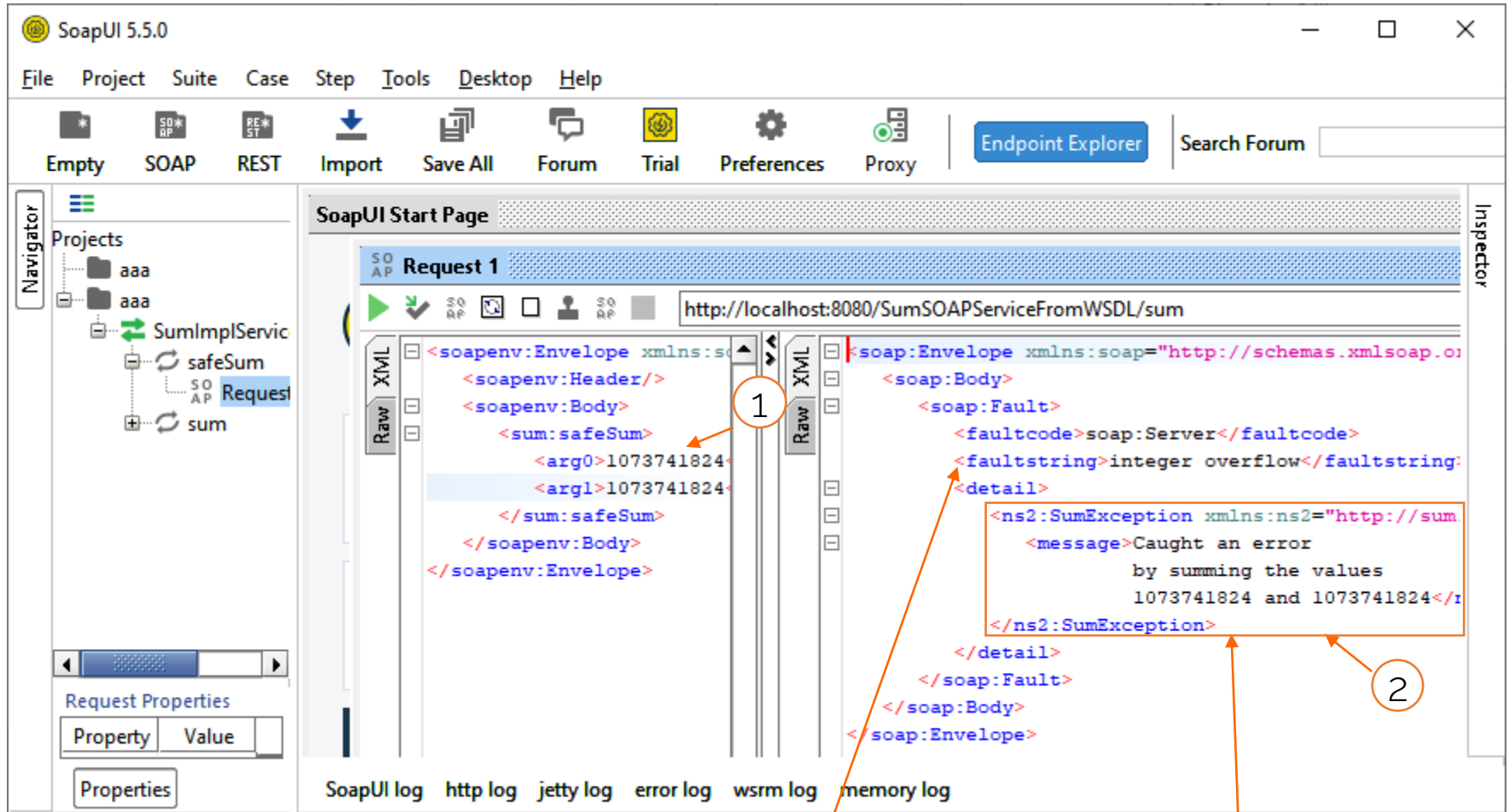
```
42 <xs:element minOccurs="0" name="sum" type="xs:string"/>
43 </xs:sequence>
44 </xs:complexType>
45 </xs:schema>
46 </wsdl:types>
47 <wsdl:message name="safeSum">
48   <wsdl:part element="tns:safeSum" name="parameters">
49   </wsdl:part>
50 </wsdl:message>
51 <wsdl:message name="SumException_Exception">
52   <wsdl:part element="tns:SumException" name="SumException_Exception">
53   </wsdl:part>
54 </wsdl:message>
55 <wsdl:message name="sum">
56   <wsdl:part element="tns:sum" name="parameters">
```

20/03/2026

Marco Autili, University of L'Aquila

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Modeled fault



```
throw new SumException_Exception(e.getMessage(), sumException);
```

More on WSDL Fault Messages...

```
SumException_Exception.java  ✕
1
2  package it.univaq.odws.sum;
3
4  import javax.xml.ws.WebFault;
5
6
7  /**
8   * This class was generated by Apache CXF 3.3.7
9   * 2020-09-27T16:13:52.091+02:00
10  * Generated source version: 3.3.7
11  */
12
13  @WebFault(name = "SumException", targetNamespace = "http://sum.odws.univaq.it/")
14  public class SumException_Exception extends Exception {
15
16      private it.univaq.odws.sum.SumException sumException;
17
18      public SumException_Exception() {
19          super();
20      }
21
22      public SumException_Exception(String message) {
23          super(message);
24      }
25
26      public SumException_Exception(String message, java.lang.Throwable cause) {
27          super(message, cause);
28      }
29
30      public SumException_Exception(String message, it.univaq.odws.sum.SumException sumException) {
31          super(message);
32          this.sumException = sumException;
33      }
34
35      public SumException_Exception(String message, it.univaq.odws.sum.SumException sumException, java.lang.Throwable cause) {
36          super(message, cause);
37          this.sumException = sumException;
38      }
39
40      public it.univaq.odws.sum.SumException getFaultInfo() {
41          return this.sumException;
42      }
43  }
44
```

How to get an un-modeled fault

The screenshot shows the Eclipse IDE workspace for 'SumSOAPServiceFromWSDL'. The Project Explorer on the left displays the project structure, with 'SumImplPortImpl.java' selected under 'src/it.univaq.odws.sum' (1). The main editor shows the code for 'SumImplPortImpl.java'. The 'safeSum' method is highlighted, showing it throws 'SumException_Exception' (3). A callout box (2) points to the 'safeSum' method signature, stating: 'In this case, we do not pass SumException as second parameter'.

```
public int sum(int arg0, int arg1) {  
    LOG.info("Executing operation sum");  
    return arg0+arg1;  
}  
  
/* (non-Javadoc)  
 * @see it.univaq.odws.sum.Sum#safeSum(int arg0, int arg1)*  
 */  
public int safeSum(int arg0, int arg1) throws SumException_Exception {  
    LOG.info("Executing operation safeSum");  
    try {  
        return Math.addExact(arg0, arg1);  
    } catch (ArithmeticException e) {  
        throw new SumException_Exception(e.getMessage());  
    }  
}
```

Un-modeled fault

The screenshot displays a SOAP client interface with two panels. The left panel shows the raw XML of a request to `http://localhost:8080/SumSOAPServiceFromWSDL/sum`. The right panel shows the raw XML of the response, which is a fault. A callout box points to the `</soap:Envelope>` tag in the response, stating: "The <detail>... </detail> tag has not been generated". Below the panels, a code snippet is shown: `throw new SumException_Exception(e.getMessage());`. The bottom status bar indicates a response time of 126ms (211 bytes) and lists various headers and attachments.

Request 1

http://localhost:8080/SumSOAPServiceFromWSDL/sum

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <sum:safeSum>
      <arg0>1073741824</arg0>
      <arg1>1073741824</arg1>
    </sum:safeSum>
  </soapenv:Body>
</soapenv:Envelope>
```

```
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <soap:Fault>
      <faultcode>soap:Server</faultcode>
      <faultstring>integer overflow</faultstring>
    </soap:Fault>
  </soap:Body>
</soap:Envelope>
```

The <detail>... </detail> tag has not been generated

```
throw new SumException_Exception(e.getMessage());
```

response time: 126ms (211 bytes)

Headers (5) Attachments (0) SSL Info WSS (0) JMS (0)

References

- <https://www.eclipse.org/downloads/packages>
- <https://tomcat.apache.org/download-80.cgi>
- <http://cxf.apache.org/download.html>
- <https://www.soapui.org/downloads/soapui.html>
- https://en.wikipedia.org/wiki/Java_API_for_XML_Web_Services
- <https://javaee.github.io/tutorial/jaxws002.html>
- <http://download.oracle.com/javaee/5/tutorial/doc/bnazq.html#bnazs>